Triple-Stage Collaborative Optimization Method for Cross-Regional Data Centers Considering Optimal Coalition Matching

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***Abstract*—With the rapid advancements of artificial intelligence, cross-regional collaboration among data centers (DCs) has emerged as an effective strategy to enhance resource utilization and reduce energy consumption. Without such collaboration, DCs may experience higher operational costs and energy consumption due to imbalances in computational load and disparities in regional electricity prices. Therefore, a triple-stage collaborative optimization method for cross-regional DCs considering optimal coalition matching is proposed. In Stage 1, a potential coalition search method for DCs is proposed, and the Bron-Kerbosch pivoting algorithm is employed to identify high-quality potential coalitions under multi-dimensional complementarity. Then, an optimal collaborative operation model for DCs incorporating fine-grained cooling system modeling is established in Stage 2, aiming to jointly optimize computational load scheduling and energy management. Based on this, an optimal dynamic coalition matching mechanism for DCs is proposed in Stage 3, integrating a rolling-horizon control strategy to adaptively update coalition structures and operation strategies under varying fluctuation levels. Case studies on ten heterogeneous cross-regional DCs demonstrate that the proposed method effectively reduces energy consumption, increases coalition profits and member satisfaction, thereby enhancing both economic performance and operational stability.**

***Index Terms*—Data center, potential coalition search, collaborative optimization, dynamic coalition matching.**

I. Introduction

THE rapid development of cloud computing, the Internet of Things (IoT) and 5G technologies has significantly increased the number and scale of data centers (DCs), which play a crucial role in supporting the digital economy and societal operations. However, as DCs proliferate, their energy consumption has become a growing concern [1]. Currently, DCs account for around 1% of global electricity consumption, approximately 460 TWh annually, and are projected to exceed 1000 TWh by 2026 [2]. To meet the dual carbon targets [3], improving energy efficiency in DCs is crucial for reducing both energy consumption and carbon emissions.

The energy consumption of DCs is primarily attributed to information technology (IT) equipment, cooling systems, and power distribution systems, which account for approximately 50%, 40% and 10% of total energy consumption, respectively [4]. Power usage effectiveness (PUE) is widely recognized as a key metric for assessing energy efficiency, where lower values indicate more efficient operations [5]. In recent years, considerable research has focused on optimizing energy management and reducing PUE in DCs. For instance, dynamic voltage and frequency scaling (DVFS) technology is implemented in [6], aiming to optimize energy management and reduce PUE values. A cooling system model in [7] accounts for outdoor weather, gas supply status and load fluctuations, reducing PUE through joint optimization of computing power and energy consumption. An energy management model considering multi-coupling effects between computing load, electricity and heat is proposed in [8], reducing operational costs in demand response scenarios.

As the range of business models for DCs diversifies, the computational loads they handle are increasingly temporally flexible. This flexibility allows for load shifting, enabling coordinated optimization of computing resources and energy consumption. For example, an optimal scheduling model for DCs based on computational load migration is established in [9], introducing a time-shifting mechanism for computational tasks. A robust optimization method for low-carbon DCs is proposed in [10], addressing renewable energy uncertainty and implementing a multiple computing task scheduling mechanism. A comprehensive analysis of the coupling between computational load and energy consumption is presented in [11], proposing an energy management method that accounts for the heterogeneity of computational loads. A risk-aware scheduling method for DCs in uncertain environments is

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proposed in [12], enhancing task scheduling efficiency through distributed reinforcement learning.

While previous research mainly focuses on individual DC optimization, the increasing number and distribution of DCs now enable cross-regional computational load transfer via the Internet. This allows for load balancing between local and remote DCs and leverages regional electricity price differences for peak-valley arbitrage [13]. Specifically, when local electricity prices are high or a load exceeds capacity, offloading tasks to a remote DC with lower electricity prices or underutilized resources helps avoid network congestion, reduce operational costs, and improve resource efficiency. Additionally, the variability of renewable energy sources like wind and solar further necessitates cross-regional collaboration to balance supply and demand. Consequently, multi-DC coordination has become a key research focus. For instance, a coordinated online scheduling framework for multiple DCs with shared energy storage is proposed in [14], using a prediction-free Lyapunov optimization algorithm to optimize service management and energy efficiency in uncertain environments. A joint optimization model for multiple DCs is established in [15], featuring a two-stage rolling scheduling strategy based on model predictive control to ensure efficient cross-regional operation. In [16], a collaborative optimization model for multi-regional DCs is proposed, utilizing computational load flexibility and regional electricity price-based demand response to reduce total operating costs. Additionally, privacy protection is crucial due to information barriers between DCs. To address this, privacy-preserving methods based on distributed Markov decision processes and distributed optimization algorithms have gained increasing attention for enhancing energy efficiency while ensuring secure resource sharing and privacy protection [17], [18].

Although significant progress has been made in existing research, several areas still need further investigation: 1) Most studies assume a fixed coalition structure among cross-regional DCs, ignoring the need for coalition search in multi-agent, resource-heterogeneous scenarios. Inappropriate coalition partitioning limits optimization potential, while uniform scheduling of all members significantly increases model complexity, making it impractical for real-world applications. 2) Existing studies often oversimplify cooling system energy consumption modeling, failing to accurately capture the impact of refrigeration on computational load scheduling and energy management. This leads to a mismatch between predicted and actual energy consumption, thereby reducing operational efficiency. 3) Most research focuses on operational optimization and profit distribution within a fixed coalition, lacking robust mechanisms for optimal coalition matching and dynamic adjustment. It limits achieving both optimal operation and stable coalition dynamics under complex and uncertain conditions.

To address the aforementioned challenges, this paper proposes a triple-stage collaborative optimization method for cross-regional DCs considering optimal coalition matching, and the key contributions are summarized as follows:

1) A potential coalition search method for cross-regional DCs based on the Bron–Kerbosch pivoting algorithm is proposed. It transforms the search into a maximal clique identification problem on a complement graph, effectively identifying high-quality potential coalitions that account for heterogeneous resource characteristics and multi-scenario uncertainties. Compared to existing methods, the proposed method effectively reduces the search space and improves search efficiency.

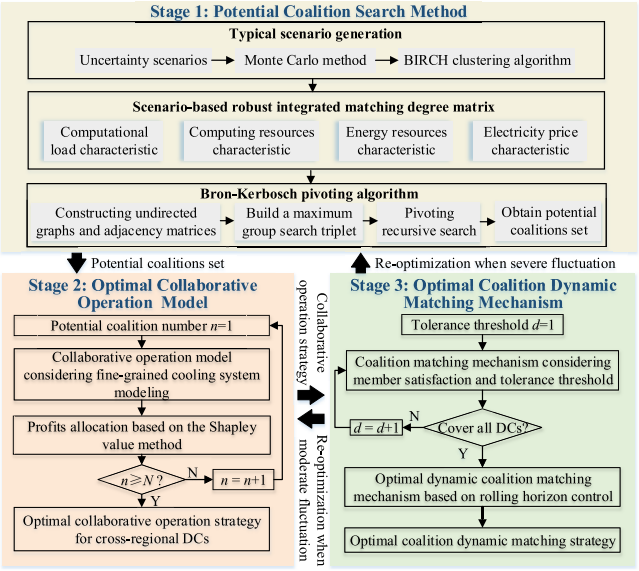


Fig. 1. Triple-stage collaborative optimization framework for cross-regional DCs considering optimal coalition matching.

2) An optimal collaborative operation model for cross-regional DCs incorporating fine-grained cooling system modeling is proposed. This model captures the coupling between cooling energy consumption and computational load scheduling, optimizing energy management within the coalition. Additionally, the Shapley value method is used for fair profit allocation. Compared to existing models, the proposed model enables more efficient scheduling and energy management, thereby reducing energy consumption and operating costs.

3) An optimal dynamic coalition matching mechanism for cross-regional DCs based on rolling horizon control is proposed. This mechanism optimizes coalition matching strategy by integrating member satisfaction and tolerance thresholds. Based on this, a rolling horizon control-driven dynamic matching mechanism optimizes coalition formation and operational strategies to maximize member satisfaction while accounting for operational uncertainties. Compared to existing approaches, the proposed mechanism better adapts to environmental changes, thereby enhancing coalition stability and profitability.

II. Triple-Stage Collaborative Optimization Framework for Cross-Regional DCs Considering Optimal Coalition Matching

In this paper, a triple-stage collaborative optimization framework for cross-regional DCs considering optimal coalition matching is proposed, which is shown in Fig. 1.

In Stage 1, a potential coalition search method for cross-regional DCs based on the Bron–Kerbosch pivoting algorithm

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is proposed. Benchmark curves for operating scenarios of computational load, renewable energy output and market price are generated using the Monte Carlo method and the balanced iterative reducing and clustering using hierarchies (BIRCH) algorithm [19]. A robust integrated matching degree matrix is then constructed to address resource heterogeneity and multi-scenario uncertainties. Using a matching degree threshold, a complement graph is created, transforming the coalition search into a maximal clique identification task. The Bron–Kerbosch pivoting algorithm is finally employed to efficiently identify high-quality potential coalitions, reducing the search space and simplifying the solution for subsequent stages. The output of Stage 1 consists solely of feasible coalition candidates and does not depend on operational or dual information from the subsequent stage. These candidates provide the structural foundation for Stage 2 and also serve as the candidate set used later in Stage 3 when severe uncertainties trigger coalition reconfiguration.

In Stage 2, an optimal collaborative operation model for cross-regional DCs is proposed. A fine-grained cooling system model is established to capture key physical characteristics, such as cold and hot aisle temperature evolution, chiller and fan power consumption. Based on this, a collaborative optimization operation model is developed to optimize energy management and computational load scheduling for all potential coalitions. Additionally, a cooperative game model using the Shapley value method ensures fair profit allocation among coalition members. Stage 2 does not alter the coalition candidates identified in Stage 1. Instead, it evaluates each candidate coalition independently and produces the corresponding optimal costs, operational decisions and profit allocations, which form the economic basis for Stage 3 to determine the final coalition configuration.

In Stage 3, an optimal dynamic coalition matching mechanism for cross-regional DCs based on rolling horizon control is proposed. Building on the optimization results from Stage 2, member satisfaction and tolerance thresholds are introduced, along with constraints such as member coverage and coalition exclusivity. The objective is to maximize overall satisfaction and determine the optimal coalition matching strategy. Real-time forecasts of uncertain variables, such as computational load and renewable generation output, are collected and updated. The rolling horizon control technique is then applied to adaptively adjust coalition structures and operational strategies based on fluctuations, ensuring stability and improving the overall benefits of collaboration. More specifically, when real-time deviations are mild, only local operational adjustments are made; when deviations are moderate, the Stage 2 optimization is re-executed for the current coalition; only when deviations become severe, the affected DCs are returned to the candidate coalition set generated in Stage 1 for re-matching, and the corresponding optimization in Stage 2 is subsequently updated.

These three stages form an integrated and coherent closed-loop framework. Stage 1 provides the feasible coalition structures, Stage 2 evaluates the operational benefits of each feasible coalition, and Stage 3 adaptively adjusts coalition structures and operational decisions based on real-time

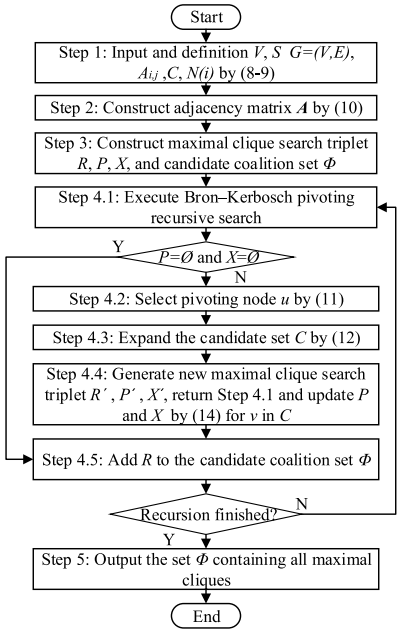


Fig. 2. Flowchart of the potential coalition search procedure based on Bron–Kerbosch pivoting algorithm.

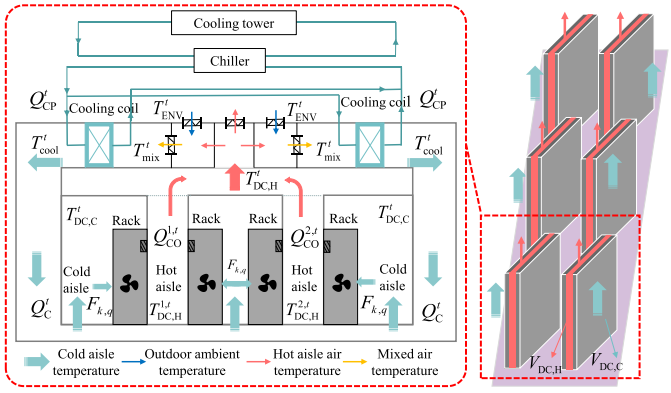


Fig. 3. Typical model of a cooling system in DC.

uncertainties while ensuring that all reconfigurations remain consistent with the feasibility rules established in Stage 1 and the operational evaluations obtained in Stage 2. The overall workflow of the three-stage framework is presented in Fig. 1, and Fig. 4 in Section V further illustrates how Stage 3 interacts with Stages 1 and 2 during dynamic reconfiguration, thereby demonstrating the complete closed-loop interaction among the three stages.

III. Stage 1: Potential Coalition Search Method for DCs Based on the Bron–Kerbosch Pivoting Algorithm

To form coalitions among N cross-regional DCs, the total number of possible coalition partitions can be expressed by the Bell number, B_N, which represents the total number of ways to partition N distinct elements into several non-empty, non-overlapping subsets. Its recursive equation is given in (1), showing that the Bell number grows exponentially with the increase in the number of DCs. For example, when N = 15,

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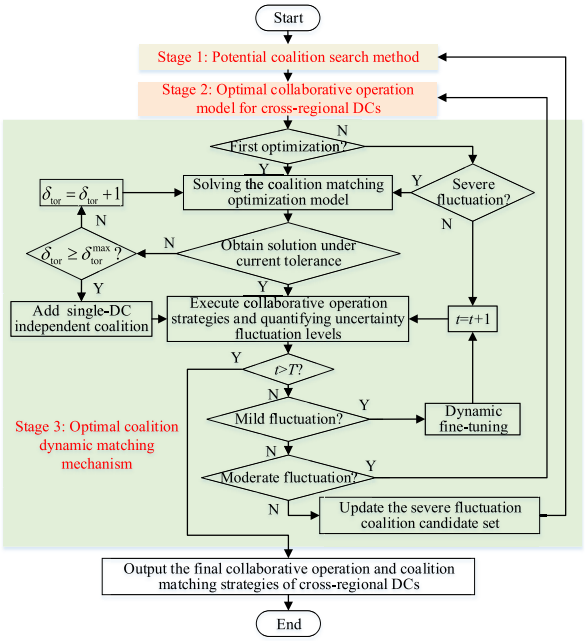


Fig. 4. Optimal dynamic coalition matching mechanism.

it exceeds 1.38×10^9, making exhaustive enumeration of all coalition structures computationally infeasible. To overcome this combinatorial explosion, the coalition formation problem is reformulated as a maximal-clique search on a matching-degree graph, where each vertex denotes a DC and edges connect pairs whose matching degree exceeds a predefined threshold. Under this formulation, a feasible coalition corresponds naturally to a maximal clique, i.e., a set of DCs in which all pairwise complementarity constraints are satisfied and no additional DC can be added without violating feasibility.

$$B_N = \sum_{k=1}^{N} B_k \left[(N-1)!\right] / \left[k!\,(N-1-k)!\right] \tag{1}$$

Based on the abovementioned graph representation, the Bron-Kerbosch pivoting algorithm is adopted to enumerate all potential coalitions. As a classical and highly efficient exact algorithm for maximal-clique enumeration, the Bron-Kerbosch pivoting algorithm iteratively expands candidate cliques and uses a pivoting strategy to prune unpromising search branches, thereby reducing recursion depth and significantly accelerating computation. This makes the algorithm particularly suitable for large-scale DC networks, where pairwise complementarity among all coalition members should be strictly enforced. By enumerating all maximal cliques, the algorithm identifies every feasible coalition while avoiding the exponential complexity associated with Bell-number enumeration. Compared with alternative methods such as clustering-based fixed-coalition methods [20], which cannot guarantee pairwise feasibility within coalitions, and grand-coalition approaches [21], which require complete information sharing among all DCs and often lead to considerable coordination difficulty and computational burden, the Bron-Kerbosch pivoting algorithm provides a balanced and scalable solution. It identifies all feasible coalitions that satisfy the complementarity criteria while preserving solution accuracy and significantly reducing the search space. These advantages make the Bron–Kerbosch pivoting algorithm particularly suitable for Stage 1 of the proposed framework.

Therefore, this paper constructs a robust integrated matching-degree matrix that incorporates forecasting uncertainties, and applies the Bron–Kerbosch pivoting algorithm to efficiently identify all high-quality potential coalitions.

A. Robust Integrated Matching Degree Quantification for Cross-Regional DCs Coalitions

Since DCs generally exhibit forecasting uncertainties in aspects such as computational load, renewable energy output and electricity market price, these factors should be fully considered during Stage 1. For N data centers, typical scenario sets of computational load curves, renewable energy output curves, and market price curves with $N \times N_S$ samples are generated using the Monte Carlo method and the BIRCH clustering algorithm. For any pair of data centers (i, j) under any scenario $s \in N_S$, a computational load complementarity indicator $C_{i,j}^{\mathrm{L},s}$ and its normalized value $S_{i,j}^{\mathrm{L},s}$, an energy resource complementarity indicator $C_{i,j}^{\mathrm{E},s}$ and its normalized value $S_{i,j}^{\mathrm{E},s}$, a market price complementarity indicator $C_{i,j}^{\mathrm{P},s}$ and its normalized value $S_{i,j}^{\mathrm{P},s}$, and a computing resource complementarity indicator $C_{i,j}^{\mathrm{R}}$ and its normalized value $S_{i,j}^{\mathrm{R}}$, are constructed. Since the expressions for the computational load, energy resource, and market price complementarity indicators are similar, taking computational load as an example, its expression is represented by (2). The expression of the computing resource complementarity indicator and its normalized value is represented by (3). A larger complementarity indicator indicates a higher degree of matching between DC i and DC j.

$$\begin{cases} C_{i,j}^{\mathrm{L},s} = \left\| \boldsymbol{I}_i^{\mathrm{L},s} - \boldsymbol{I}_j^{\mathrm{L},s} \right\|_2 = \sqrt{\sum_{t=1}^{T} \left(I_i^{\mathrm{L},s,t} - I_j^{\mathrm{L},s,t} \right)^2} \\ S_{i,j}^{\mathrm{L},s} = C_{i,j}^{\mathrm{L},s} / \left[\max_{\forall i,j} \left(C_{i,j}^{\mathrm{L},s} \right) \right], S_{i,i}^{\mathrm{L},s} = 0 \end{cases} \tag{2}$$

$$\begin{cases} C_{i,j}^{\mathrm{R}} = 1 - \dfrac{\boldsymbol{p}_i \cdot \boldsymbol{p}_j}{\left\| \boldsymbol{p}_i \right\|_2 \left\| \boldsymbol{p}_j \right\|_2}, \boldsymbol{p}_i = \left[p_i^{\mathrm{CPU}}, p_i^{\mathrm{GPU}}, p_i^{\mathrm{FPGA}}, p_i^{\mathrm{TPU}} \right] \\ S_{i,j}^{\mathrm{R}} = C_{i,j}^{\mathrm{R}}, S_{i,i}^{\mathrm{R}} = 0 \end{cases} \tag{3}$$

where T is the number of scheduling periods; $\boldsymbol{I}_i^{\mathrm{L},s}$ is the set of computational load energy consumption for DC i under scenario s; $I_i^{\mathrm{L},s,t}$ is the computational load energy consumption of DC i at time t under scenario s; $\boldsymbol{p}_i$ is the set of IT resource proportions owned by DC i, including central processing unit (CPU), graphics processing unit (GPU), field-programmable gate array (FPGA) and tensor processing unit (TPU).

Based on the four complementarity indicators, the weighted Euclidean distance $d_{i,j}^s$ of coalition (i, j) in scenario s is calculated by (4). Then, $d_{i,j}^s$ is normalized to obtain $S_{i,j}^s$, as represented by (5). Next, all $S_{i,j}^s$ are utilized to form the integrated matching degree matrix $\boldsymbol{S}^s$, as represented by (6). Finally, the robust mean degree $\tilde{S}_{i,j}(\forall i, j)$ is obtained by aggregating results across all scenarios, thereby constructing

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the final robust integrated matching degree matrix $\boldsymbol{S}$, as represented by (7). It should be noted that w^{L}, w^{E}, w^{P} and w^{R} are weight coefficients of $S_{i,j}^{\mathrm{L},s}$, $S_{i,j}^{\mathrm{E},s}$, $S_{i,j}^{\mathrm{P},s}$ and $S_{i,j}^{\mathrm{R}}$, respectively. Taking w^{L} as an example, it can be obtained by combining objective weight $w_{\mathrm{obj}}^{\mathrm{L}}$ and subjective weight $w_{\mathrm{sub}}^{\mathrm{L}}$, as shown in (8). Among them, $w_{\mathrm{obj}}^{\mathrm{L}}$ can be obtained by the entropy weight method [22], and $w_{\mathrm{sub}}^{\mathrm{L}}$ can be obtained by the AHP method [23].

$$d_{i,j}^{s}=\sqrt{w^{\mathrm{L}}\left(S_{i,j}^{\mathrm{L},s}\right)^{2}+w^{\mathrm{E}}\left(S_{i,j}^{\mathrm{E},s}\right)^{2}+w^{\mathrm{P}}\left(S_{i,j}^{\mathrm{P},s}\right)^{2}+w^{\mathrm{R}}\left(S_{i,j}^{\mathrm{R}}\right)^{2}} \tag{4}$$

$$S_{i,j}^{s}=(d_{i,j}^{s}-d_{\min}^{s})/(d_{\max}^{s}-d_{\min}^{s}),S_{i,j}=S_{j,i},S_{i,i}=1 \tag{5}$$

$$\boldsymbol{S}^{s}=\left[S_{i,1}^{s}S_{i,2}^{s}\ldots S_{i,N}^{s}\right]_{i=1,\,2,\ldots,N} \tag{6}$$

$$\tilde{S}_{i,j}=\sum_{\forall s\in N_S}S_{i,j}^{s}/N_S,\boldsymbol{S}=[\tilde{S}_{i,j}],\forall i,j\in N \tag{7}$$

$$w^{\mathrm{L}}=\alpha w_{\mathrm{obj}}^{\mathrm{L}}+(1-\alpha)\,w_{\mathrm{sub}}^{\mathrm{L}} \tag{8}$$

where $d_{\max}^{s}$ and $d_{\min}^{s}$ are the maximum and minimum values of all $d_{i,j}^{s}$ in scenario s, respectively; α is an auxiliary weight coefficient.

It should be noted that the four complementarity indicators are normalized using equations (2), (3), (5) and (7). This normalization process ensures that all complementarity indicators are placed on a comparable scale, thereby eliminating the influence of differing units and magnitudes.

B. Potential Coalition Search Procedure for Cross-Regional DCs Based on the Bron–Kerbosch Pivoting Algorithm

Based on the robust integrated matching degree matrix $\boldsymbol{S}$ obtained in Section III-A, this section introduces the application of the Bron–Kerbosch pivoting algorithm to identify high-quality potential coalitions of cross-regional DCs. The detailed procedure is outlined as follows, and the corresponding flowchart is shown in Fig. 2.

Step 1: Input and definitions. The input parameters include the set of DCs V ={1, 2,..., N}, the robust integrated matching degree matrix S, and the matching degree threshold $\theta\in[0,1]$. An undirected graph is defined as $G=(V,E)$, where the edge set E is given as (9). A clique C is defined as a subset of V, i.e., $C\subseteq V$. For any two DCs $i,j\in C$, the adjacency state $A_{i,j}=1$. A maximal clique is defined as a clique C to which no additional node $i\in V\backslash C$ can be added without violating the clique property. For $\forall i\in V$, its neighbor set $N(i)$ is defined as (10).

$$E=\{(i,j)|i<j,\tilde{S}_{i,j}\geq\theta\} \tag{9}$$

$$N(i)=\{\forall j\in V|A_{i,j}=1\} \tag{10}$$

where θ is a matching degree threshold.

Step 2: Constructing the undirected graph and adjacency matrix. Based on the previous definitions, the adjacency matrix $\boldsymbol{A}$ is constructed as shown in (11), enabling quick determination of connectivity and neighbor set extraction for each node. In the undirected graph $G=(V,E)$, each edge represents a pair of DCs with an integrated matching degree greater than or equal to the given threshold θ, indicating potential for coalition formation.

$$\boldsymbol{A}=\left[A_{i,j}\right]^{N\times N},$$
$$A_{i,j}=\begin{cases}1, & \tilde{S}_{i,j}\geq\theta\\ 0, & \text{otherwize}\end{cases},A_{i,i}=0,A_{i,j}=A_{j,i} \tag{11}$$

Step 3: Constructing the maximal clique search triplet. The Bron–Kerbosch pivoting algorithm recursively constructs a maximal clique using three dynamically updated sets: (1) Current clique $R\subseteq V$ represents the set of nodes currently selected. (2) Candidate set $P\subseteq V$ represents the set of candidate nodes that can be added to R at the current state. (3) Excluded set $X\subseteq V$ represents the set of nodes that will not be considered in this branch. All cliques that satisfy the maximal clique property will be collected into the candidate coalition set Φ. The initial state of the triplet is defined as ($R=\emptyset$, $P=V$, $X=\emptyset$, $\Phi=\emptyset$).

Step 4: Execute Bron–Kerbosch pivoting recursive search. This procedure consists of the following five sub-steps.

Step 4.1: Termination check and addition of maximal cliques. The current state is checked to see if P= $\emptyset$ and X= $\emptyset$. If true, this indicates that the current clique R is a maximal clique, meaning it cannot be expanded further and is not a subset of any larger clique. In this case, the clique R is added to the candidate coalition set $\Phi=\Phi\cup\{R\}$, and the procedure moves to *Step 4.5*. Otherwise, turn to *Step 4.2*.

Step 4.2: Selection of pivoting node. In the Bron–Kerbosch pivoting algorithm, each recursion involves multiple candidate nodes $i\in P$ that can be used to expand the current clique R. The role of the pivoting node u is to optimize the pruning process by prioritizing the skipping of parts that are strongly associated with certain known nodes, thereby reducing unnecessary recursion branches and accelerating the search procedure. Thus, given the current triplet (R, P, X), the pivoting node u is selected from set $P\cup X$, as it has the highest number of neighboring nodes within the candidate set P, as shown in (12).

$$u=\arg\max_{\forall i\in P\cup X}|P\cap N(i)| \tag{12}$$

Step 4.3: Expanding the candidate set. Based on the pivoting node u, all candidate nodes that are not connected to u form the expanded set C, as represented by (13). If C is empty, it indicates that there are no suitable new nodes to add to the current clique, and the procedure moves to *Step 4.5*. Otherwise, turn to *Step 4.4*.

$$C=P\backslash N(u) \tag{13}$$

Step 4.4: Recursive search for each expanded candidate node $v\in C$. For each node v in the set C, a new state is constructed according to (14), and the recursive search procedure from *Step 4.1* is repeated with the updated triplet (R', P', X'). If the sub-recursion satisfies the conditions of *Step 4.1*, the corresponding maximal clique is added to Φ, and the procedure returns to this step. Regardless of whether a new clique is found, the current set should be updated according to (15) upon returning to this step to avoid redundant recursion.

$$R'=R\cup\{v\},P'=P\cap N(v),X'=X\cap N(v) \tag{14}$$

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$$P = P\backslash\{v\}, X = X \cup \{v\} \tag{15}$$

Step 4.5: Check if the current recursive call is the outermost call. If it is, turn to *Step 5*; otherwise, the current recursive call ends and the procedure returns to the previous level of recursion.

Step 5: End of potential coalition search procedure. Output the set Φ containing all maximal cliques (i.e., potential coalitions) that satisfy the matching degree threshold.

IV. Stage 2: Optimal Collaborative Operation Model for Cross-Regional DCs Incorporating Fine-Grained Cooling System Modeling

This section proposes an optimal collaborative operation model for cross-regional DCs incorporating fine-grained cooling system modeling, aiming to maximize the overall operational benefits of coalitions and ensure fair profit allocation. The model requires only key summary indicators (e.g., load demand curves, equipment efficiency and renewable forecasts) from coalition members, without accessing sensitive customer or business data. All data is exchanged under a non-disclosure agreement. For higher privacy needs, federated optimization or distributed computing methods can be employed.

A. Fine-Grained Cooling System Modeling of DCs

Cooling systems account for a substantial portion of total energy consumption in modern DCs, making accurate cooling modeling essential for optimizing cross-regional collaborative operation. However, traditional cooling-energy estimation methods, such as PUE-based [24] or COP-based [8] linear conversion, approximate cooling demand as a fixed proportion of IT load. Although computationally convenient, these simplified models overlook key thermodynamic and airflow mechanisms, including cold/hot aisle separation, supply-return air temperature variation, air-mixing behavior and mixed-air mass flow. As a result, such approximations may yield biased cooling-energy estimates and lead to suboptimal or even misleading operational decisions in coalition scheduling. To overcome these limitations, a fine-grained cooling system model is employed in this paper. This modeling approach captures the physical coupling between IT workload adjustment and cooling-system behavior, enabling a more realistic estimation of cooling energy consumption across DCs.

Building on this modeling framework, Fig. 3 illustrates the airflow structure of a typical air-cooled DC. The cooling system introduces conditioned air into the cold aisle, and the heat generated by IT equipment is discharged through the hot aisle. Fans drive the circulation of cold and hot air streams, facilitating heat exchange and maintaining appropriate rack-level operating temperatures. The key technical parameters shown in Fig. 3 are introduced: Q^t_{CP} represents the cooling power supplied by the chiller; T^t_{ENV} is the outdoor ambient temperature; $T^{k,t}_{\mathrm{DC,H}}$ is the temperature of the k-type hot aisle at time t; $T^t_{\mathrm{DC,H}}$ is the total return heat temperature of the DC from the various hot aisles after mixing at time t; T^t_{mix} is the mixed air temperature of T^t_{ENV} and $T^t_{\mathrm{DC,H}}$; T^t_{cool} is the cold air temperature injected into the cold aisle after refrigeration at time t; $T^t_{\mathrm{DC,C}}$ is the cold aisle temperature at time t.

1) Fresh-Air Process Modeling of Cooling Systems: The cooling power Q^t_{CP} is determined by the mass flow rate of the cooled air and the temperature difference before and after cooling, as given in (16)-(18).

$$Q^t_{\mathrm{CP}} = m^t c_{\mathrm{air}}(T^t_{\mathrm{mix}} - T^t_{\mathrm{cool}}) \tag{16}$$

$$m^t = \rho_{\mathrm{air}} \sum\nolimits_{k=1}^{N_k} M^{\mathrm{fan}}_k F^t_k \tag{17}$$

$$F^t_k = \sum\nolimits_{q=1}^{N_q} a_{k,t,q} F_{k,q}, \sum\nolimits_{q=1}^{N_q} a_{k,t,q} = 1 \tag{18}$$

where m^t is the total mass of the mixed air obtained by combining the return air from the DC and the outdoor fresh air at time t; c_{air} is the specific heat capacity of air; ρ_{air} is the air density; M^{fan}_k is the total number of fans in the k-type hot aisle (i.e., the hot aisle serving k-type servers); F^t_k is the airflow rate driven by a single fan in the k-type hot aisle at time t; N_q is the number of fan speed levels; $a_{k,t,q}$ is the fan speed state of the k-type hot aisle at time t; $F_{k,q}$ is the airflow rate corresponding to the q-th speed level of a fan in the k-type hot aisle.

The mixed air results from proportionally blending the hot aisle return air and the outdoor fresh air. Accordingly, the mixed-air temperature T^t_{mix} is represented by (19). The total return heat temperature $T^t_{\mathrm{DC,H}}$ depends on the return-air temperatures $T^{k,t}_{\mathrm{DC,H}}$ of different hot aisles and the fan speed states F^t_k, as given in (20).

$$T^t_{\mathrm{mix}} = \delta_{\mathrm{mix}} T^t_{\mathrm{ENV}} + (1 - \delta_{\mathrm{mix}}) T^t_{\mathrm{DC,H}} \tag{19}$$

$$T^t_{\mathrm{DC,H}} = \left(\sum\nolimits_{k=1}^{N_k} M^{\mathrm{fan}}_k F^t_k T^{k,t}_{\mathrm{DC,H}}\right) / \left(\sum\nolimits_{k=1}^{N_k} M^{\mathrm{fan}}_k F^t_k\right) \tag{20}$$

where δ_{mix} is the mixing ratio coefficient between the outdoor ambient temperature and the total return heat temperature.

2) Physical Variation Process Modeling of Hot and Cold Aisles in DC Cooling Systems: According to the law of energy conservation, nearly all electrical energy consumed by IT equipment is converted into heat, meaning the thermal power released by IT equipment equals the electrical power consumed by the servers, as shown in (21). The change rate of internal energy in the cold aisle is the difference between the heat exchanged with the outdoor environment and the cooling power supplied, so the thermal balance equations for the cold and hot aisles are given by (22)-(23), respectively. Additionally, the temperatures of both aisles should stay within their rated ranges to ensure proper cooling system operation, as shown in (24)-(25).

$$Q^{k,t}_{\mathrm{H}} = M^k_{\mathrm{ser}} P^{k,t}_{\mathrm{ser}} \tag{21}$$

$$c_{\mathrm{air}}\rho_{\mathrm{air}} V_{\mathrm{DC,C}} \Delta T^t_{\mathrm{DC,C}} = \left(T^t_{\mathrm{ENV}} - T^t_{\mathrm{DC,C}}\right)/R_{\mathrm{DC}} - Q^t_{\mathrm{C}} \tag{22}$$

$$c_{\mathrm{air}}\rho_{\mathrm{air}} V_{\mathrm{DC,H}} \Delta T^t_{\mathrm{DC,H}} = Q^{k,t}_{\mathrm{H}} - Q^{k,t}_{\mathrm{CO}} \tag{23}$$

$$T^{\min}_{\mathrm{DC,H}} \le T^{k,t}_{\mathrm{DC,H}} \le T^{\max}_{\mathrm{DC,H}} \tag{24}$$

$$T^{\min}_{\mathrm{DC,C}} \le T^t_{\mathrm{DC,C}} \le T^{\max}_{\mathrm{DC,C}} \tag{25}$$

where $Q^{k,t}_{\mathrm{H}}$ is the total heat generation power of the k-type hot aisle during time t; M^k_{ser} is the number of servers in the k-type hot aisle; $P^{k,t}_{\mathrm{ser}}$ is the active power consumed by each server in the k-type hot aisle at time t; $V_{\mathrm{DC,C}}$ and $V_{\mathrm{DC,H}}$ are the

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volumes of the cold and hot aisles, respectively; $\Delta T^t_{\mathrm{DC,C}}$ and $\Delta T^t_{\mathrm{DC,H}}$ are the temperature variations of the cold and hot aisles during time t, respectively, $\Delta T^t_{\mathrm{DC,C}} = T^{t+1}_{\mathrm{DC,C}} - T^t_{\mathrm{DC,C}}$, $\Delta T^{k,t}_{\mathrm{DC,H}} = T^{k,t+1}_{\mathrm{DC,H}} - T^{k,t}_{\mathrm{DC,H}}$; R_{DC} is the thermal resistance between the DC and the outdoor environment; $T^{\max}_{\mathrm{DC,H}}$ and $T^{\min}_{\mathrm{DC,H}}$ are the upper and lower bounds of $T^{k,t}_{\mathrm{DC,H}}$, respectively; $T^{\max}_{\mathrm{DC,C}}$ and $T^{\min}_{\mathrm{DC,C}}$ are the upper and lower bounds of $T^t_{\mathrm{DC,C}}$, respectively; Q^t_{C} and $Q^{k,t}_{\mathrm{CO}}$ are the cooling power supplied to the cold aisle and the k-type hot aisle at time t, respectively, whose expressions are represented by (26)-(27).

$$Q^t_{\mathrm{C}} = m^t c_{\mathrm{air}}(T^t_{\mathrm{DC,C}} - T^t_{\mathrm{cool}}) \tag{26}$$

$$Q^{k,t}_{\mathrm{CO}} = m^t_k c_{\mathrm{air}}(T^{k,t}_{\mathrm{DC,H}} - T^t_{\mathrm{DC,C}}) \tag{27}$$

3) Energy Consumption Modeling of Cooling Systems: The energy consumption of the cooling system, i.e., P^t_{CO}, arises from the chiller, fans, and auxiliary cooling devices, as represented by (28). The energy consumption of chiller P^t_{CP} is characterized by its coefficient of performance (COP) in (29). The fan operates at discrete speed levels in practice, which can be selected and adjusted within several given levels. Therefore, the fans' energy consumption P^t_{fan} can be linearly approximated and expressed as (30). P^t_{a} represents the power consumption of auxiliary cooling equipment, including water pumps and cooling towers, and is set as a constant in this paper.

$$P^t_{\mathrm{CO}} = P^t_{\mathrm{CP}} + P^t_{\mathrm{fan}} + P^t_{\mathrm{a}} \tag{28}$$

$$P^t_{\mathrm{CP}} = Q^t_{\mathrm{CP}}/\eta_{\mathrm{COP}} \tag{29}$$

$$P^t_{\mathrm{fan}} = \gamma^{\mathrm{F}}\rho^3_{\mathrm{air}} \sum\nolimits_{k=1}^{N_k} \sum\nolimits_{q=1}^{N_q} M^{\mathrm{fan}}_k a_{k,t,q}(F_{k,q})^3 \tag{30}$$

where η_{COP} is the chiller's COP; γ^{F} is the fan power coefficient.

B. Optimal Collaborative Operation Model for DCs

For any potential coalition $W \subseteq \Phi$ obtained in the first stage, the optimization objective is to maximize the net operational benefits of all coalition members. As shown in (31)-(32), this objective is defined as the computational load service revenue minus the total operational costs, including IT energy consumption, cooling-system energy consumption, and the task-migration cost associated with inter-DC workload transfers. The stochastic programming method is utilized in this paper to address multiple uncertainties during the operation. The total power consumption of the i-th DC under scenario s, i.e., $P^{i,x,t}_{\mathrm{E}}$, consists of the server power consumption $P^{i,x,t}_{\mathrm{IT}}$, the cooling system power consumption $P^{i,x,t}_{\mathrm{CO}}$, and the auxiliary power supply consumption $P^{i,t}_{\mathrm{A}}$, as represented by (33). $P^{i,x,t}_{\mathrm{IT}}$ is mainly related to server type and CPU operating frequency [6]. In this paper, the DVFS technology is introduced to construct an approximate expression for server energy consumption [8], as represented by (34)-(35). The expression for $P^{i,x,t}_{\mathrm{CO}}$ is similar to that in (28)-(30). The value of $P^{i,t}_{\mathrm{A}}$ can be set as a constant. It should be noted that the task-migration cost is proportional to the data size of the transmitted request [25], and DCs typically operate under bandwidth-package or flat-rate agreements, so the marginal cost of additional data transfer is negligible [26]. Thus, communication overhead is implicitly included within the task-migration cost and does not require separate modeling.

$$\max_{B^{i,x}} \sum\nolimits_{x=1}^{X} \sum\nolimits_{i=1}^{N^W_{\mathrm{DC}}} B^{i,x} \tag{31}$$

$$= \sum\nolimits_{t=1}^{T} \left(r^{i,t}_{\mathrm{Comp}} R^{i,x,t}_{\mathrm{Local,C}} - c^i_{\mathrm{Trans}} \sum D^{i,j,t}_{\mathrm{r,trans}} - c^{i,x,t}_{\mathrm{E}} P^{i,x,t}_{\mathrm{E}} \right) \Delta t \tag{32}$$

$$P^{i,x,t}_{\mathrm{E}} = P^{i,x,t}_{\mathrm{IT}} + P^{i,x,t}_{\mathrm{CO}} + P^{i,t}_{\mathrm{A}} \tag{33}$$

$$P^{i,x,t}_{\mathrm{IT}} = \sum\nolimits_{k=1}^{N_k} M^k_{\mathrm{ser}} P^{i,x,t}_{\mathrm{ser}} = \sum\nolimits_{k=1}^{N_k} M^k_{\mathrm{ser}} \left(P^{i,x,k,t}_{\mathrm{CPU}} + P^{i,x,k,t}_{\mathrm{fixed}} \right) \tag{34}$$

$$P^{i,x,k,t}_{\mathrm{CPU}} = \gamma_{\mathrm{CPU}} \sum\nolimits_{s=1}^{N_s} b^{i,x,k,t,s}_{\mathrm{CPU}} (f^{i,x,k,s}_{\mathrm{CPU}})^2,$$
$$s.t. \sum\nolimits_{s=1}^{N_s} b^{i,x,k,t,s}_{\mathrm{CPU}} = 1 \tag{35}$$

where X is the number of uncertainty scenarios; N^W_{DC} is the number of DCs in the potential coalition W; $B^{i,x}$ is the net operating profit of the i-th DC; $r^{i,t}_{\mathrm{Comp}}$ is the unit price of computing service revenue for the i-th DC at time t; c^i_{Trans} is the unit price of the cross-regional load migration cost; $D^{i,j,t}_{\mathrm{trans}}$ is the volume of batch-processing computational load migrated from the i-th DC to the j-th DC at time t; $R^{i,x,t}_{\mathrm{Local,C}}$ is the amount of locally processed computational load of the i-th DC at time t; $c^{i,x,t}_{\mathrm{E}}$ is the unit energy consumption cost of the i-th DC at time t; Δt is the scheduling time interval; $P^{i,x,k,t}_{\mathrm{CPU}}$ is the active power consumption of the CPU in each k-type server at time t; $P^{i,x,k,t}_{\mathrm{fixed}}$ is the baseline idle power of each k-type server; γ_{CPU} is the CPU energy consumption coefficient; N_s is the number of CPU frequency levels; $b^{i,x,k,t,s}_{\mathrm{CPU}}$ is the state variable indicating the selected CPU frequency level; $f^{i,x,k,s}_{\mathrm{CPU}}$ is the s-th CPU operating frequency.

The operational constraints of the proposed model include cooling system operation constraints (16)-(30), spatio-temporal scheduling constraints for computational loads (36)-(40), and power balance constraint (41). For simplicity, the superscript x will not be displayed in the following text. Equation (36) represents the locally processed computational load including both real-time and batch computational loads; equation (37) defines the relationship constraint between CPU utilization and the locally processed computational load; equation (38) defines the relationship constraint between the locally processed and cross-regional transferred real-time computational loads; equation (39) defines the relationship constraint between the remaining batch computational load and the unfinished computational load processed locally within the data center; equation (40) represents the delay characteristics constraint of the batch computational load. The power balance constraint for the DC is represented by (41).

$$R^{i,t}_{\mathrm{Local,C}} = x^{i,t}_{\mathrm{r}} + x^{i,t}_{\mathrm{b}} \tag{36}$$

$$R^{i,t}_{\mathrm{Local,C}} = \sum\nolimits_{k=1}^{N_k} M^k_{\mathrm{ser}} u^{i,k,t}_{\mathrm{CPU}} C^{\mathrm{Per},i}_{\mathrm{CPU}} \left(\sum\nolimits_{s=1}^{N_s} b^{i,k,t,s}_{\mathrm{CPU}} f^{i,k,s}_{\mathrm{CPU}} \right),$$
$$s.t. u^{i,k,t}_{\mathrm{CPU}} \le u^{i,\max}_{\mathrm{CPU}} \tag{37}$$

$$x^{i,t}_{\mathrm{r}} = D^{i,t}_{\mathrm{r}} - \sum_{\forall j \ne i} D^{i,j,t}_{\mathrm{r,trans}} + \sum_{\forall i \ne j} D^{j,i,t}_{\mathrm{r,trans}},$$
$$s.t. 0 \le \sum_{\forall j \ne i} D^{i,j,t}_{\mathrm{r,trans}} \le \delta^{i,j,t} D^{i,t}_{\mathrm{r}} \tag{38}$$

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$$x_{\mathrm{b}}^{i,t} = Q_{\mathrm{b}}^{i,t-1} + D_{\mathrm{b}}^{i,t} - Q_{\mathrm{b}}^{i,t},$$
$$s.t. Q_{\mathrm{b}}^{i,t} \geq 0, Q_{\mathrm{b}}^{i,0} = 0, Q_{\mathrm{b}}^{i,T} = 0 \quad (39)$$

$$\sum\nolimits_{t}^{t+T_{\mathrm{b}}^{\max}} x_{\mathrm{b}}^{i,t} \geq \beta_{\mathrm{b}}^{\min} \sum\nolimits_{t}^{t+T_{\mathrm{b}}^{\max}} D_{\mathrm{b}}^{i,t} \quad (40)$$

$$P_{\mathrm{grid}}^{i,t} + P_{\mathrm{RES}}^{i,t} = P_{\mathrm{IT}}^{i,t} + P_{\mathrm{CO}}^{i,t} + P_{\mathrm{A}}^{i,t} \quad (41)$$

where $x_{\mathrm{r}}^{i,t}$ and $x_{\mathrm{b}}^{i,t}$ are the locally processed volumes of real-time and batch-processing computational loads of the i-th DC at time t, respectively; $u_{\mathrm{CPU}}^{i,k,t}$ is the CPU utilization of the k-type servers in the i-th DC at time t; $C_{\mathrm{CPU}}^{\mathrm{Per},i}$ is the computational performance coefficient of the servers in the i-th DC (in cores/GHz); $u_{\mathrm{CPU}}^{i,\max}$ is the secure-operating threshold of CPU utilization for the servers in the i-th DC; $D_{\mathrm{r}}^{i,t}$ and $D_{\mathrm{b}}^{i,t}$ are the arrival volumes of real-time and batch-processing computational loads of the i-th DC at time t, respectively; $Q_{\mathrm{b}}^{i,t}$ is the remaining workload of the batch-processing computational load of the i-th DC at time t; $\beta_{\mathrm{b}}^{\min}$ is the lower bound on the proportion of deferred batch-processing computational load that should be completed within a specified time window; $T_{\mathrm{b}}^{\max}$ is the maximum allowable delay time for batch-processing computational load; $P_{\mathrm{grid}}^{i,t}$ and $P_{\mathrm{RES}}^{i,t}$ are the grid-purchased power and distributed renewable generation output of the i-th DC at time t, respectively.

It should be noted that equations (16), (20), (26) and (27) contain non-convex terms, which involve the product of binary and continuous variables. This makes it difficult to directly construct a convex optimization model and solve it using commercial solvers. To address this, equivalent linear transformations of these equations are conducted using the big-M method, which is a standard technique for linearizing such non-convex constraints [27]. The method introduces auxiliary variables and linear constraints to reformulate the original non-linear terms into mathematically equivalent linear forms. This approach ensures the accuracy of the model and guarantees that the optimality of the solution is preserved [28]. The detailed process of this linearization is provided in equations (42)-(55).

More specifically, auxiliary variables $b_{k,t,q}^{\mathrm{mix}} = a_{k,t,q} T_{\mathrm{mix}}^{t}$ and $b_{k,t,q}^{\mathrm{mix}} = a_{k,t,q} T_{\mathrm{mix}}^{t}$ are introduced to linearize equation (16), resulting in the linearized form shown in equations (42)-(46). Auxiliary variables $b_{k,t,q}^{\mathrm{DC,H}} = a_{k,t,q} T_{t}^{\mathrm{DC,H}}$ and $b_{k,t,q}^{\mathrm{H}} = a_{k,t,q} T_{k,t}^{\mathrm{DC,H}}$ are introduced to linearize equation (20), and the resulting model is shown in equations (47)-(51). Similarly, equations (26) and (27) also contain non-convex and nonlinear terms. Therefore, the auxiliary variable $b_{k,t,q}^{\mathrm{C}} = a_{k,t,q} T_{\mathrm{DC,C}}^{t}$ is introduced to transform equation (26) into equations (52)-(54), and equation (27) is transformed into equation (55).

$$Q_{\mathrm{CP}}^{t} = \rho_{\mathrm{air}} c_{\mathrm{air}} \sum\nolimits_{k=1}^{N_k} M_{k}^{\mathrm{fan}} \sum\nolimits_{q=1}^{N_q} F_{k,q} (b_{k,t,q}^{\mathrm{mix}} - b_{k,t,q}^{\mathrm{cool}}) \quad (42)$$

$$T_{\mathrm{mix}}^{t} - (1 - a_{k,t,q}) M_{\mathrm{SLC}} \leq b_{k,t,q}^{\mathrm{mix}} \leq a_{k,t,q} M_{\mathrm{SLC}} \quad (43)$$

$$0 \leq b_{k,t,q}^{\mathrm{mix}} \leq T_{\mathrm{mix}}^{t} + (1 - a_{k,t,q}) M_{\mathrm{SLC}} \quad (44)$$

$$T_{\mathrm{cool}}^{t} - (1 - a_{k,t,q}) M_{\mathrm{SLC}} \leq b_{k,t,q}^{\mathrm{cool}} \leq a_{k,t,q} M_{\mathrm{SLC}} \quad (45)$$

$$0 \leq b_{k,t,q}^{\mathrm{cool}} \leq T_{\mathrm{cool}}^{t} + (1 - a_{k,t,q}) M_{\mathrm{SLC}} \quad (46)$$

$$\rho \sum\nolimits_{k=1}^{N_k} M_{k}^{\mathrm{fan}} \sum\nolimits_{q=1}^{N_q} F_{k,q} b_{k,t,q}^{\mathrm{DC,H}} = \rho \sum\nolimits_{k=1}^{N_k} M_{k}^{\mathrm{fan}} \sum\nolimits_{q=1}^{N_q} F_{k,q} b_{k,t,q}^{\mathrm{H}} \quad (47)$$

$$T_{\mathrm{DC,H}}^{t} - (1 - a_{k,t,q}) M_{\mathrm{SLC}} \leq b_{k,t,q}^{\mathrm{DC,H}} \leq a_{k,t,q} M_{\mathrm{SLC}} \quad (48)$$

$$0 \leq b_{k,t,q}^{\mathrm{DC,H}} \leq T_{\mathrm{DC,H}}^{t} + (1 - a_{k,t,q}) M_{\mathrm{SLC}} \quad (49)$$

$$T_{\mathrm{DC,H}}^{k,t} - (1 - a_{k,t,q}) M_{\mathrm{SLC}} \leq b_{k,t,q}^{\mathrm{H}} \leq a_{k,t,q} M_{\mathrm{SLC}} \quad (50)$$

$$0 \leq b_{k,t,q}^{\mathrm{H}} \leq T_{\mathrm{DC,H}}^{k,t} + (1 - a_{k,t,q}) M_{\mathrm{SLC}} \quad (51)$$

$$Q_{t}^{\mathrm{C}} = \rho c_{\mathrm{air}} \sum\nolimits_{k=1}^{N_k} M_{k}^{\mathrm{fan}} \sum\nolimits_{q=1}^{N_q} F_{k,q} (b_{k,t,q}^{\mathrm{C}} - b_{k,t,q}^{\mathrm{cool}}) \quad (52)$$

$$T_{\mathrm{DC,C}}^{t} - (1 - a_{k,t,q}) M_{\mathrm{SLC}} \leq b_{k,t,q}^{\mathrm{C}} \leq a_{k,t,q} M_{\mathrm{SLC}} \quad (53)$$

$$0 \leq b_{k,t,q}^{\mathrm{C}} \leq T_{\mathrm{DC,C}}^{t} + (1 - a_{k,t,q}) M_{\mathrm{SLC}} \quad (54)$$

$$Q_{\mathrm{CO}}^{k,t} = \rho c_{\mathrm{air}} M_{k}^{\mathrm{fan}} \sum\nolimits_{q=1}^{N_q} F_{k,q} (b_{k,t,q}^{\mathrm{H}} - b_{k,t,q}^{\mathrm{C}}) \quad (55)$$

where M_{SLC} is a sufficiently large constant.

Thus, the optimal collaborative operation model for cross-regional DCs incorporating fine-grained cooling system modeling is formulated as a mixed-integer linear programming (MILP) model. It should be noted that the stability of a coalition requires fair benefit distribution. Therefore, the Shapley value method is employed in this paper to construct a cooperative game model, and the profit allocation for the i-th DC in a potential coalition W can be calculated using (56), thereby achieving fair profit distribution among the coalition members.

$$\varphi^{W,i} = \sum_{\forall S \subseteq W \setminus \{i\}} [|S|!(|W| - |S| - 1)!/|W|!] [v(S \cup \{i\}) - v(S)] \quad (56)$$

where $v(S) = \sum_{\forall i \in S} B^{W,i}$ is the value function of any subset S of a potential coalition W, $B^{W,i}$ is the net operational profit of the i-th DC in W; $|S|$ is the number of elements in the subset S.

V. Stage 3: Optimal Dynamic Coalition Matching Mechanism for DCs Based on Rolling Horizon Control

In the potential coalition candidate set $\Phi = \{s_1, ..., s_m, ..., s_M\}$ (M is the total number of potential coalitions), there is often member overlap between candidate coalitions, i.e., $\exists \{s_m, s_n\} \in \Phi, s_m \cap s_n \neq \phi$, which are referred to as conflicting coalitions. If multiple potential coalitions with the highest individual profits are selected directly, member duplication may prevent these coalitions from coexisting simultaneously. Additionally, the uncertainty in the predictions of computational load and renewable energy output can impact the operational profits of the coalitions, affecting the optimality and stability of the coalition matching results. To address these issues, an optimal dynamic coalition matching mechanism for cross-regional DCs based on rolling horizon control is proposed.

A. Coalition Matching Optimization Model Incorporating Member Satisfaction and Tolerance Threshold

In the collaborative operation of cross-regional DCs, the satisfaction of coalition members with the profit allocation directly impacts the optimal coalition matching results. The satisfaction of a coalition member reflects their level of satisfaction with the profit allocation within a specific coalition,

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compared to the profit they would receive in all potential coalitions they are part of. Set the profit allocated to the i-th DC in a candidate coalition s_m ($\forall s_m \in \Phi$) be $R^i_{s_m}$, then the satisfaction indicator $U^i_{s_m}$ for coalition s_m is defined as (57). In (57), $U^i_{s_m}$ is normalized to the range [1, 0], with higher values indicating greater attractiveness of coalition s_m for the i-th DC. It should be noted that the matching degree in Stage 1 is a technical complementarity metric used before coalition formation, while the satisfaction level in Stage 3 reflects individual rationality and determines whether coalition members perceive the outcome as acceptable. Therefore, the matching degree in Stage 1 guides the structural search for potential coalitions, and the satisfaction level in Stage 3 selects the final coalition by evaluating economic fairness and stability.

$$U^i_{s_m} = \left(R^i_{s_m} - R^i_{\min}\right) / \left(R^i_{\max} - R^i_{\min}\right) \tag{57}$$

where $R^i_{\max}$ and $R^i_{\min}$ are the maximum and minimum profit allocated to the i-th DC among all potential coalitions that include itself, respectively.

Then, a tolerance threshold coefficient δ_{tor} is introduced to reflect the minimum acceptable satisfaction ranking of the i-th DC for a potential coalition. A coalition s_m is considered acceptable to the i-th DC only if it satisfies the tolerance threshold condition shown in (58). Based on this, the set of all acceptable coalitions for the i-th DC under the tolerance threshold coefficient δ_{tor} is defined as $\boldsymbol{Y}^i$, which is represented by (59); the set of all candidate coalitions that satisfy the tolerance threshold constraint is defined as $\boldsymbol{Y}$, which is represented by (60). To ensure model feasibility amid coalition conflicts or tolerance violations, a fallback mechanism for single DC coalitions is introduced, as shown in (61). More specifically, when the tolerance reaches the maximum acceptable threshold $\delta^{\max}_{\text{tor}}$, if there are still no acceptable coalitions for some DCs, each DC will be added as a candidate coalition operating independently to the new candidate coalitions set $\boldsymbol{Y'}$, ensuring the model always has a feasible solution.

$$\text{rank}(U^i_{s_m}) \le \delta_{\text{tor}} \tag{58}$$

$$\boldsymbol{Y}^i = \left\{\forall s_m \in \Phi | \text{rank}(U^i_{s_m}) \le \delta_{\text{tor}}\right\} \tag{59}$$

$$\boldsymbol{Y} = \boldsymbol{Y}^1 \cup \boldsymbol{Y}^2 \cup \ldots \cup \boldsymbol{Y}^N \tag{60}$$

$$\boldsymbol{Y'} = \boldsymbol{Y} \cup \{1, 2, \ldots, N\}, \text{ when } \delta_{\text{tor}} \ge \delta^{\max}_{\text{tor}} \tag{61}$$

where $\text{rank}(U^i_{s_m})$ is the ranking of the satisfaction indicator $U^i_{s_m}$ of the i-th DC for all coalitions containing it, with the rankings sorted in descending order.

Finally, the coalition matching optimization model is formulated as (62)-(65) to determine an optimal and no-overlapping coalition Φ^{opt} that maximizes member satisfaction. Specifically, the objective function is defined in (62) as maximizing the total profit of all successfully matched coalitions, namely, the sum of all selected coalitions' profits minus the penalty cost for unselected DCs. The constraint of the fallback mechanism for single DC coalitions is imposed in (63). The requirement that each DC can join only one coalition in the final matching results is ensured in (64). Equation (65) ensures that the satisfaction of each DC should exceed the tolerance threshold in any selected coalition.

$$\max \sum_{\forall s_m \in \boldsymbol{Y'}} \sum_{\forall i \in s_m} x_m R^i_{s_m} - \delta_{\text{pen}} \sum_{\forall i \in N} \left(1 - y^i\right) \tag{62}$$

$$y^i = \sum_{\forall s_m | i \in s_m} x_m \le 1, i = 1, 2, \ldots, N \tag{63}$$

$$x_m + x_n \le 1, \forall s_m, s_n \subseteq \boldsymbol{Y'}, m \ne n \tag{64}$$

$$\text{rank}(U^i_{s_m}) \le \delta_{\text{tor}}, \forall i \in s_m, \forall s_m \subseteq \boldsymbol{Y'} \tag{65}$$

where x_m is an auxiliary binary variable, which equals 1 if coalition s_m is selected as one of the optimal coalitions; y^i is an auxiliary binary variable, which equals 1 if at least one potential coalition containing the i-th DC is included in the optimal coalitions; δ_{pen} is the penalty coefficient for unselected DCs.

B. Optimal Dynamic Coalition Matching Mechanism Based on Rolling Horizon Control

During cross-regional DC operations, various real-world disturbances, such as stochastic fluctuations in computational load, intermittent renewable energy output and short-term electricity price volatility, may cause deviations from the planned operational schedule. These disturbances can affect coalition-level operating costs and compromise the optimality and stability of the coalition structure. To address these practical operational disruptions, this paper incorporates an optimal dynamic coalition matching mechanism based on rolling-horizon control. As illustrated in Fig. 4, the mechanism continuously evaluates the deviation between real-time observations and forecasted conditions and triggers different levels of corrective actions accordingly. Mild deviations are handled through local operational adjustments without altering the coalition structure; moderate deviations activate re-optimization of the operational decisions within the existing coalition; severe deviations prompt affected coalitions to re-enter the candidate pool for re-matching. This hierarchical and event-driven adjustment strategy enhances the robustness of the proposed framework, ensuring that coalition formation and operation remain feasible, stable and economically efficient under practical uncertainty. The detailed procedure is as follows.

Step 1: Parameter initialization. The potential coalition search results from Stage 1, and the collaborative operation and profit allocation strategies from Stage 2 are input, together with the rolling horizon control window length and technical parameters such as uncertainty fluctuation thresholds ε_1 and ε_2. Set $\delta_{\text{tor}} = 1$ and $t = 1$.

Step 2: Candidate coalition selection. For each DC, the candidate coalitions set $\boldsymbol{Y}^i$ ranked within the top δ_{tor} in terms of satisfaction is obtained according to (59), and the current candidate coalition set $\boldsymbol{Y}$ is determined according to (60).

Step 3: Solving the optimal coalition matching results. The coalition matching optimization model described in (62)-(65) is solved to determine whether a coalition matching solution covering all members exists. If such a solution exists, the optimal coalition matching result Φ^{opt} is obtained and turns to *Step 4*. Otherwise, it is checked whether $\delta_{\text{tor}} \ge \delta^{\max}_{\text{tor}}$. If

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true, the candidate coalition set $\boldsymbol{Y}$ is expanded by adding single-DC independent coalitions according to (61) and then form the candidate set $\boldsymbol{Y}'$, after which the optimal coalition matching result Φ^{opt} is obtained and turns to *Step 4*; otherwise, $\delta_{\text{tor}} = \delta_{\text{tor}} + 1$, and return to *Step 2*.

Step 4: Executing collaborative operation strategies and quantifying uncertainty fluctuation levels. For each coalition $w \subseteq \Phi^{\text{opt}}$, the collaborative operation strategy at the current time t_1 is executed. For the period from t_1 to the end of the scheduling horizon, forecasts of computational load and renewable generation output of each DC are collected and updated. The uncertainty fluctuation level d^w_{unc} of each coalition w is then calculated according to (66).

$$d^w_{\text{unc}} = \sum_{\forall i \in w} \sum_{t=t_1}^{T} \left[\left| D^{i,t}_{\text{real}} - D^{i,t}_{\text{pre}} \right| / D^{i,t}_{\text{pre}} + \left| P^{i,t}_{\text{RES,real}} - P^{i,t}_{\text{RES,pre}} \right| / P^{i,t}_{\text{RES,pre}} \right] \quad (66)$$

where $D^{i,t}_{\text{real}}$ and $D^{i,t}_{\text{pre}}$ are the actual and forecasted values of the computational load of the DC, respectively; $P^{i,t}_{\text{RES,real}}$ and $P^{i,t}_{\text{RES,pre}}$ are the actual and forecasted values of the renewable energy output of the DC, respectively.

Step 5: Hierarchical rolling horizon control of the triple-stage model. According to the uncertainty fluctuation indicator of each coalition, differentiated processing is carried out under the following three fluctuation levels:

1) Mild fluctuation ($d^w_{\text{unc}} < \varepsilon_1$). The coalition structure and collaborative operation strategy remain unchanged, with each DC adjusting grid power purchases, renewable energy curtailment, and local load scheduling to mitigate uncertainty.

2) Moderate fluctuation ($\varepsilon_1 \le d^w_{\text{unc}} < \varepsilon_2$). The coalition structure remains unchanged, but the collaborative strategies of cross-regional DCs are updated by re-solving the Stage 2 model within each coalition.

3) Severe fluctuation ($\varepsilon_1 \le d^w_{\text{unc}} < \varepsilon_2$). Coalitions with severe fluctuations are added to the candidate set, and the full triple-stage optimization process (Stages 1-3) is re-executed to update both the collaborative operation and optimal coalition matching strategies.

Step 6: Termination check of rolling horizon control. Determine whether $t > T$. If true, the rolling horizon control process is completed, and the final collaborative operation and coalition matching strategies of cross-regional DCs are output. Otherwise, $t = t + 1$, and turn to *Step 4*.

It is worth noting that the fluctuation classification in Stage 3 is conducted at the coalition level rather than at the level of individual DCs. Although different members may experience different degrees of forecasting deviations in renewable output or computing load, the dynamic adjustment mechanism evaluates the aggregated fluctuation indicator of the coalition as a whole. The subsequent decision, i.e., whether to perform minor operational adjustments, re-optimize the coalition's joint operation, or reintroduce the coalition into the matching process, is therefore based on the overall deviation level. This coalition-level treatment avoids excessive sensitivity to isolated extreme disturbances experienced by single members, thereby preventing unnecessary coalition restructuring and ensuring stability and practical applicability in real-time operation.

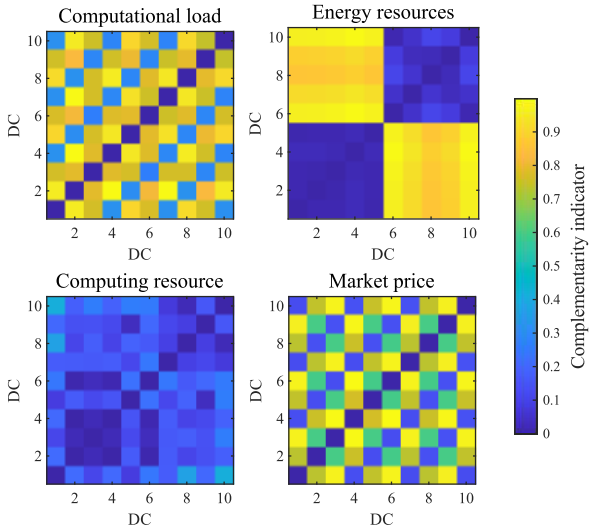


Fig. 5. Normalized complementarity heatmaps of 10 cross-regional DCs.

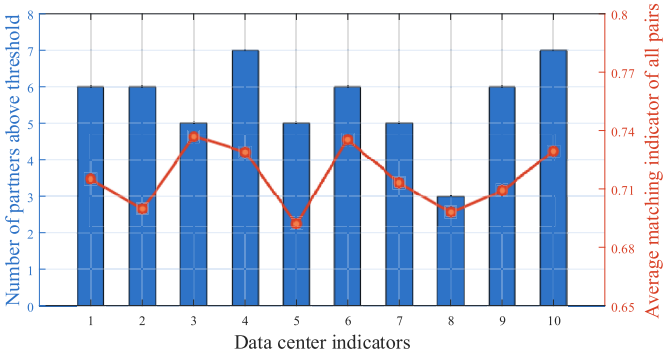


Fig. 6. Number of high-match coalitions and their corresponding average complementarity for each DC.

Additionally, the multiple DCs considered are owned by the same enterprise. However, despite being part of the same organization, a coalition matching algorithm is still necessary due to concerns regarding data privacy and internal governance. Specifically, although all DCs belong to the same enterprise, they typically serve different internal business units or external clients, which necessitates a certain level of data isolation [17]. If coalition matching were to be centrally managed by the enterprise, privacy protection issues could arise, particularly when large volumes of data are involved across multiple departments, as these privacy concerns would become more complex. Therefore, this paper allows each DC to independently select the coalition it wishes to join. Since the results of coalition matching generally involve smaller groups, it is feasible for data to be uploaded to the cloud for centralized processing within small-scale coalitions. This method enhances practical feasibility, as data sharing is confined to the members of the small-scale coalition, thereby reducing the risk of privacy breaches.

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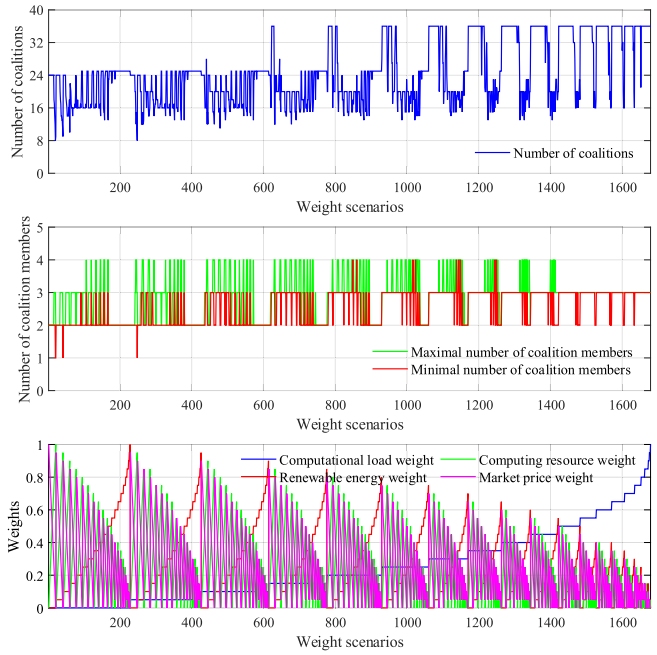


Fig. 7. Sensitivity analysis of weight coefficients.

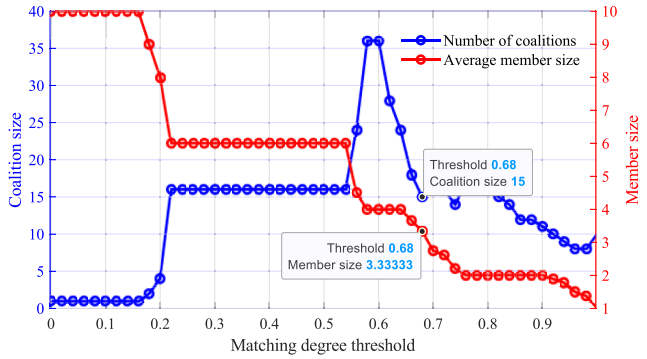


Fig. 8. Sensitivity analysis of matching degree threshold.

TABLE I

POTENTIAL COALITIONS SEARCH RESULTS AND THEIR MATCHING DEGREE CHARACTERISTICS

Coalition members	Average matching degree	Range of matching degree	Coalition members	Average matching degree	Range of matching degree
3,7	0.961	/	5,7,9	0.819	0.725,0.913
1,3,8	0.808	0.732,0.891	1,2,6,10	0.818	0.684,0.984
1,3,10	0.804	0.695,0.986	1,2,9,10	0.803	0.684,0.942
2,6,7	0.842	0.739,0.928	2,4,6,10	0.823	0.691,0.999
2,7,9	0.827	0.725,0.928	2,4,9,10	0.808	0.691,0.958
3,4,8	0.819	0.745,0.909	4,5,6,10	0.820	0.689,0.999
3,4,10	0.807	0.691,0.986	4,5,9,10	0.803	0.689,0.958
5,6,7	0.837	0.739,0.913			

VI. CASE STUDIES

Case studies are conducted on ten cross-regional DCs with heterogeneous resource characteristics to verify the effectiveness and advantages of the proposed method. The benchmark data used to construct operating scenarios are derived from real-world datasets. Specifically, computational load traces are taken from Alibaba's 2018 cluster dataset [29], while renewable energy output and electricity market price profiles are generated based on representative historical photovoltaic/wind data and time-of-use pricing in Fujian Province. Representative baseline operating curves for these variables are extracted using the BIRCH algorithm, after which Monte Carlo sampling is applied to introduce stochastic fluctuations around the clustered baselines [19]. The resulting baseline operating curves are shown in Fig. 12 of the Appendix, capturing both typical operating characteristics and realistic variability. The main technical parameters are summarized in Table IX [30].

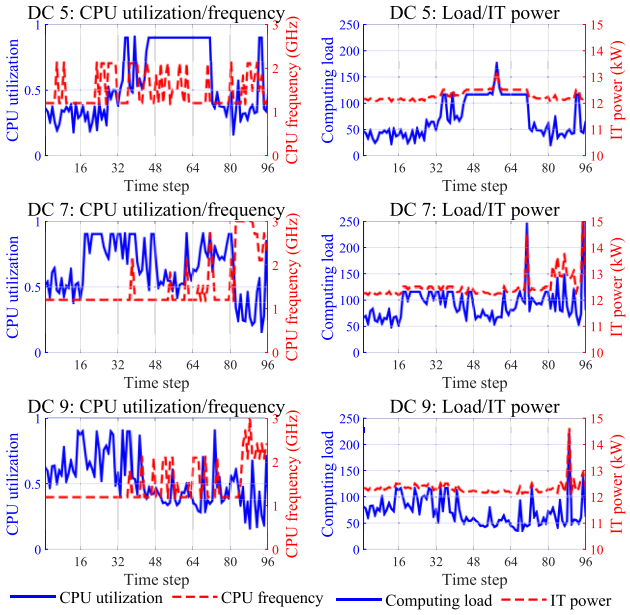


Fig. 9. CPU utilization, CPU frequency, computational load, and IT energy consumption curves of coalition {5,7,9}.

TABLE II

SOLVING RESULTS COMPARISON AMONG M-KM, M-SGC AND THE PROPOSED MODEL

Indicators	M-KM [31]	M-SGC [21]	The proposed method
Coalition structure	{1,10}/{2,5,8}/{3,4,6,7,9}	{1,2,…,10}	{3,7}/{2,4,6}/{1,9,10}/{5,8}
Total profits/CNY	17098	18128	18126
Satisfaction level/%	{46,64}/{42,66,50}/{55,63,64,64,44}	/	{97,74}/{78,68,77}/{74/87/66}/{100/50}
Practical feasibility	better	worse	better

TABLE III

COMPARISON OF THE SOLVING RESULTS BETWEEN THE PROPOSED BRON–KERBOSCH PIVOTING ALGORITHM AND THE TRADITIONAL BRON–KERBOSCH ALGORITHM

N	Conventional Bron–Kerbosch algorithm [32]			Proposed Bron–Kerbosch pivoting algorithm		
	Recursion number	Maximal clique number	Solving time	Recursion number	Maximal clique number	Solving time
10	71	15	0.28 s	31	15	0.12 s
50	1979	1009	1.05 s	1044	1009	0.76 s
100	11703	6261	5.9 s	6557	6261	4.7 s
500	345346	265985	699 s	292735	265985	568 s

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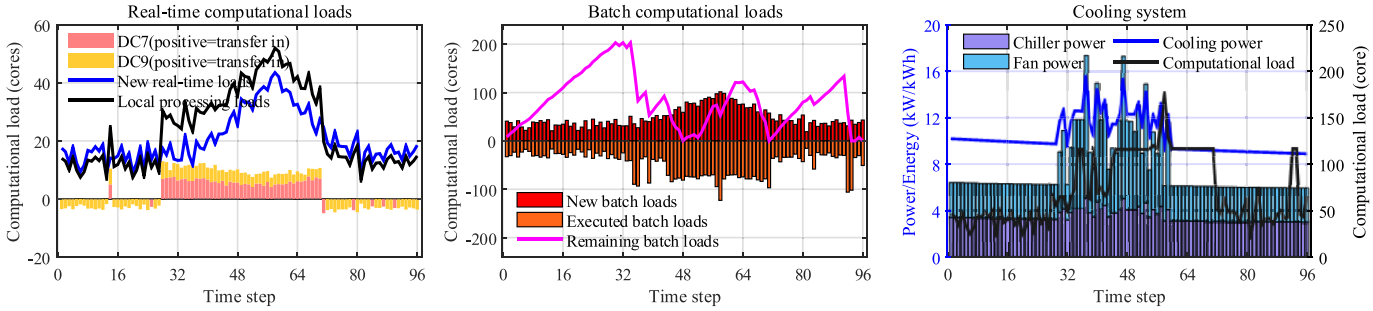


Fig. 10. Coordinating computational load and cooling system scheduling strategies of DC 5.

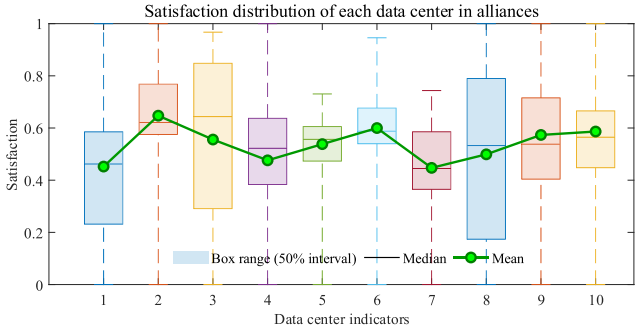


Fig. 11. Boxplots of the satisfaction levels of each DC across 155 candidate coalitions.

TABLE IV

COMPARISON OF OPTIMIZATION RESULTS AMONG M-WLT, M-PUE, M-COP, M-COP-F AND THE PROPOSED MODEL

Model	Chiller energy/ kWh	Fans energy/ kWh	IT energy/ kWh	Renewable energy/ kWh	Purchased electricity /kWh	Energy cost/ CNY	PUE
M-WLT [33]	3326	3967	11864	13767	5390	3835	1.62
M-PUE [24]	4939	4944	11945	13767	8060	6142	1.83
M-COP [8]	4765	5124	11974	13767	8096	6220	1.82
M-COP-F [34]	3549	3678	12042	13767	5502	4092	1.60
Proposed model	3283	3794	11872	13767	5181	3706	1.59

TABLE V

OPTIMAL COALITION MATCHING AND PROFIT ALLOCATION RESULTS (TOLERANCE THRESHOLD = 8)

Indicators	Optimal coalition matching result			
	{3, 7}	{2, 4, 6}	{1, 9, 10}	{5, 8}
Total profits/CNY	3717	5114	6392	2903
Profit allocation/CNY	1238/2479	2130/1154/1830	2117/2461/1813	1197/1706
Tolerance rank	2/4	5/8/6	3/3/8	1/5
Satisfaction level/%	97/74	78/68/77	74/87/66	100/50

Based on the generated multi-scenario datasets, resource complementarity is then evaluated across four key dimensions: computational load, energy resources, computing resources, and market environment. The normalized complementarity heatmaps are presented in Fig. 5, where larger values represent stronger collaboration potential between DCs.

A. Analysis of Potential Coalition Search Results for Cross-Regional DCs in Stage 1

According to the normalized complementarity heatmaps shown in Fig. 5, Fig. 6 further shows the number of high-matching coalitions that can be formed for each DC (with a matching degree threshold of 0.68) and their corresponding average matching degree. All DCs, except DC8, which has only three high-matching partners, can form coalitions with five to seven other DCs. The average matching degree ranges from 0.69 to 0.74, indicating strong multidimensional collaboration potential. Additionally, Table I lists 15 potential coalitions and their matching degree characteristics, obtained using the Bron–Kerbosch pivoting algorithm. The results show that the average matching degree of three-member coalitions exceeds 0.807, and four-member coalitions range from 0.803 to 0.823, demonstrating the method's effectiveness in preserving multidimensional complementarity. In addition, all coalitions in Table I are maximal cliques, meaning adding any additional DC would reduce the matching degree below the threshold, ensuring stable and non-expandable coalition structures. In summary, the proposed method generates stable coalition configurations with strong complementarity, providing a robust foundation for subsequent optimization in Stage 2.

To evaluate the effectiveness of the proposed potential coalition search method in Stage 1, it is compared with two baseline methods: 1) a fixed-coalition method based on the K-medoids clustering algorithm (M-KM) [31]; 2) a global optimization method that treats all DCs as a single grand coalition (M-SGC) [21]. The comparison results are presented in Table II. It can be seen from Table II that the proposed model demonstrates clear advantages over the M-KM. More specifically, the total profit obtained by the M-KM is 17098 CNY, which is lower than the 18126 CNY achieved by the proposed model. In addition, the profit allocation results show that the satisfaction levels produced by the clustering method are considerably lower: the mean satisfaction of M-KM is 5.0% lower and the minimum satisfaction is 21.3% lower compared with those of the proposed model, indicating weaker fairness and poorer coalition stability across members. The M-SGC achieves a slightly higher total profit of 18128 CNY, since all DCs jointly optimize operational decisions. However, this method requires computing profit allocation by enumerating all 2^N-1 sub-coalitions for Shapley-value-based

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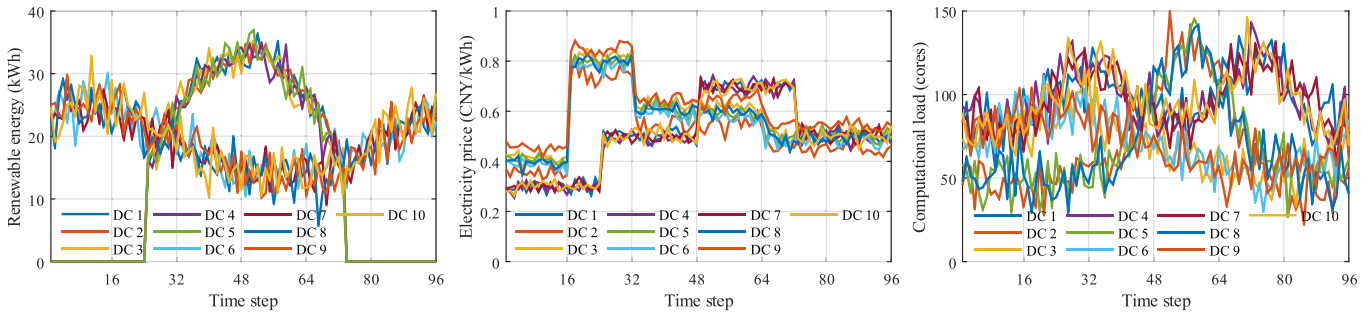


Fig. 12. Benchmark curve for operating scenarios of DCs.

profit distribution, resulting in extremely high computational complexity that is impractical when the number of DCs grows. Moreover, forming a grand coalition entails full information sharing among all DCs, which may violate data privacy requirements and operational boundaries commonly present in multi-department or multi-region DC ecosystems. In contrast, both the proposed method and the M-KM only require data exchange within small coalitions, greatly improving practical feasibility.

Overall, although the M-SGC yields marginally higher profits, its excessive computational burden and full-information requirement significantly limit its applicability. The M-KM is computationally efficient but suffers from lower profit and DCs' satisfaction. By comparison, the proposed model achieves near-optimal total profit while ensuring notably higher satisfaction levels and better practical feasibility, thereby providing a more balanced and practically viable potential coalition formation strategy for cross-regional DCs.

To further evaluate the computational performance of the Bron-Kerbosch pivoting algorithm in terms of potential coalition searches, its performance is compared with that of the conventional Bron-Kerbosch algorithm under different DC scales [32], as shown in Table III. For small-scale scenarios such as $N = 10$ or 50, both algorithms show minor differences in recursion count and computation time. However, as the number of DCs increases, the recursion depth and computation time of the conventional algorithm grow significantly. When $N = 500$, the conventional algorithm requires 345346 recursions and 699 seconds, while the proposed pivoting algorithm reduces the recursion count by 22.98% and the computation time by 18.74%. Notably, both algorithms yield identical maximal clique results, indicating that the proposed method retains solution accuracy while enhancing efficiency. Overall, the Bron-Kerbosch pivoting algorithm exhibits better computational efficiency and scalability in large-scale scenarios.

To analyze the impact of varying weight scenarios on coalition formation, the results for the number of coalitions and the member distribution in each coalition are shown in Fig. 7. The horizontal axis represents the number of weight scenario combinations, derived by varying the four weights of matching indicators. Each weight increases by 0.05, following the order of computational load, renewable output, computing resources, and market price. Since the weight changes for the last two indicators occur too rapidly, we focus on analyzing the first two indicators.

As seen in the upper subfigure, the number of potential coalitions is low when the weight for computational load matching is small. As the weight increases, the number of coalitions steadily rises, reaching a peak of 36. This indicates that prioritizing computational load matching increases coalition formation. In contrast, increasing the weight for renewable generation output reduces the number of potential coalitions. In the middle subfigure, as the computational load weight increases, coalition sizes become more consistent, with earlier scenarios showing a mix of 1-4 members, gradually stabilizing at 2-3 members. This demonstrates that prioritizing computational load matching leads to more balanced and stable coalition sizes. A similar trend is observed with renewable energy weight, where coalition sizes stabilize, though with fewer fluctuations. Overall, the sensitivity analysis shows that varying the four weight indicators impacts coalition number and member structure.

To better analyze the impact of the matching degree threshold on coalition structure, Fig. 8 shows the variations in the number of potential coalitions and the average member size under different matching degree thresholds. It can be seen from Fig. 8 that the number of potential coalitions initially increases and then decreases as the threshold rises. At lower thresholds (below 0.22), most DCs are grouped into a few large coalitions, resulting in a relatively low number of potential coalitions. As the threshold increases, the number of potential coalitions steadily rises, peaking at 36 when the threshold reaches approximately 0.6. However, as the threshold continues to increase, the number of potential coalitions starts to decrease. This is because higher thresholds reduce compatibility between DCs, making it increasingly difficult to form viable coalitions. The red curve, which represents the average coalition member size, follows a distinct pattern. When the threshold is lower than 0.16, all 10 DCs are combined into a single large coalition, resulting in a large member size. As the threshold increases, member sizes gradually decrease, reflecting the diminishing compatibility between DCs as the threshold becomes more stringent. Notably, when the threshold approaches 1, each DC operates independently, as no pair of DCs can achieve a matching degree of 1.

Ultimately, a threshold of 0.68 was selected in this paper, which results in 15 potential coalitions with an average of

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3.33 members per coalition. This threshold provides a balance between the number of potential coalitions and the stability of the coalition member structures, while also reducing the computational burden for the subsequent coalition profit allocation process.

B. Analysis of Collaborative Operation Strategy for Cross-Regional DCs in Stage 2

To analyze collaborative operation strategies within DCs, the coalition {5, 7, 9} is taken as an example. The curves of CPU utilization, CPU frequency, computational load, and IT energy consumption for these DCs are shown in Fig. 9.

It can be seen from Fig. 12 in Appendix and Fig. 9 that the local electricity price of DC7 exceeds 0.7 CNY/kWh during time periods 50 to 60, while the wind power output remains low. Consequently, part of DC7's computational load is offloaded to other coalition members, reducing the local load from a maximum of 116.6 cores to an average of 80.2 cores, lowering average CPU utilization to 61.8%, and average CPU frequency to 1.31 GHz. As a result, the average IT energy consumption drops to 12.34 kW. Similarly, during periods 35-60, DC5 experiences increased computational demand, lower electricity prices, and rising photovoltaic output. Therefore, its average locally processed computational load rises from 61.4 to 121.5 cores, its average CPU utilization increases from 47.4% to 90.0%, and its average IT energy consumption grows to 12.6 kW. In summary, the proposed model allows DCs to adjust CPU utilization, frequency, and energy consumption in response to electricity prices and computational load, reducing energy costs and enhancing coalition profits.

The coordinating computational load and cooling system scheduling strategies of DC5 over the entire horizon are shown in Fig. 10. It can be seen from the first subfigure of Fig. 10 that DC5 serves as the primary receiver of real-time computational load. This is attributed to its local electricity price remaining below 0.6 CNY/kWh and relatively high PV generation during time periods 28 to 70. Consequently, it receives up to 13 cores per time step of net inbound load. During other periods, increased local electricity prices and reduced PV output trigger part of DC5's real-time load transfer to DC7 and DC9, enabling optimal cross-regional load allocation. The second subfigure exhibits significant temporal flexibility of the batch computational load. For example, during periods 15–34, high electricity prices and insufficient PV output cause some newly generated batch loads to be deferred, accumulating 203.3 cores peak load. Subsequently, falling electricity prices and increased PV output accelerate batch task execution of DC5 during time periods 35 to 70, reducing the remaining batch load to an average of 67.3 cores. The third subfigure further highlights the coordinated optimization between the cooling system and the computational load of DC5. For instance, during time periods 30 to 70, the peak and average computational loads reach 175.0 cores and 105.4 cores, respectively. Correspondingly, the average combined power demand of the chiller and fans increases to 9.7 kW, with the cooling power frequently exceeding 12.0 kW and peaking at 15.4 kW, ensuring adequate thermal management under high-load conditions. In summary, the proposed method coordinates spatio-temporal coordination of computational load scheduling based on local electricity price levels and energy resource availability, while dynamically matching cooling power to computational load variations, thereby reducing overall energy consumption and enhancing operational economics.

To evaluate the advantages of the proposed collaborative operation model for cross-regional DCs in Stage 2, it is compared with four baseline counterparts: 1) the model without considering cross-regional computational load transfer (M-WLT) [33]; 2) the model using the PUE metric to approximate cooling energy consumption (M-PUE) [24]; 3) the model computing cooling energy consumption via coefficient of performance-based conversion (M-COP) [8]; 4) the model combining COP-based cooling estimation with fans' energy consumption (M-COP-F) [34]. The comparison results are presented in Table IV.

It can be seen from Table IV that the proposed model outperforms the other models in key metrics, including chiller energy consumption, fan energy consumption, purchased electricity, total energy cost, and PUE. More specifically, compared with M-WLT, the proposed model enables spatial transfer of computational loads among coalition members, significantly reducing peak loads at certain DCs. This optimizes the utilization of both computing and energy resources within the coalition, resulting in a 3.9% reduction in purchased electricity and a 3.4% reduction in total energy cost. The M-PUE and M-COP estimate cooling-system energy consumption by converting IT energy into cooling demand using historical PUE and COP values, respectively. These coarse approximations introduce significant estimation errors because they neglect the underlying thermal dynamics of the cooling infrastructure. Consequently, both models tend to generate overly optimistic scheduling decisions, which may elevate server inlet temperatures in actual operation, induce additional cooling demand, and ultimately increase total energy cost. As a result, the total energy costs of M-PUE and M-COP rise to 6142 CNY and 6320 CNY, respectively, substantially higher than those of the other models. The M-COP-F enhances estimation accuracy by explicitly modeling fan energy consumption, thereby reducing total energy cost by 33.4% and 34.2% relative to M-PUE and M-COP. However, it does not model key thermodynamic factors such as cold and hot aisle separation, air-mixing temperature, and mixed-air mass flow, which have significant impacts on cooling efficiency. In contrast, the proposed model incorporates a detailed thermo–airflow representation of the cooling system, enabling a more accurate assessment of cooling energy requirements. This refined modeling supports more effective and targeted operational strategies, reducing total energy cost by an additional 9.4% compared with M-COP-F.

C. Analysis of Optimal Coalition Matching and Rolling Horizon Control Strategies for Cross-Regional DCs in Stage 3

To assess the effectiveness of the proposed coalition matching optimization model, Fig. 11 presents boxplots of satisfaction levels of each DC across 149 candidate coalitions, including all potential coalitions and their sub-coalitions. Due

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to significant differences in resource complementarity, the profit allocation and satisfaction of each DC vary across coalitions. Overall, DC2, DC3, and DC6 exhibit high median and mean satisfaction values, with medians above 0.6 and means above 0.5, indicating stable profit allocations. In particular, DC2's narrow interquartile range reflects its strong adaptability to coalition structures and consistent satisfaction with profit allocations. DC8 shows the largest interquartile range among all DCs, indicating the highest variability in satisfaction. Although its median satisfaction is relatively low, the long upper whisker reveals that DC8 can achieve high satisfaction in certain coalition structures. In conclusion, structural changes in cross-regional coalitions significantly affect certain DCs, emphasizing the need to evaluate satisfaction levels to maintain coalition stability.

Based on this, the optimal coalition matching and profit allocation results are presented in Table V. A tolerance threshold of 8 indicates that rank 8 is the lowest acceptable satisfaction level. As shown in Table V, the optimal coalition matching results are {3, 7}, {2, 4, 6}, {1, 9, 10} and {5, 8}, with total coalition profits of 3717 CNY, 5114 CNY, 6392 CNY and 2903 CNY, respectively. Among them, coalition {3, 7} achieves a high level of internal stability, with DCs 3 and 7 attaining satisfaction levels of 97 percent and 74 percent, respectively. Coalition {2, 4, 6} also demonstrates acceptable fairness, where three DCs obtain satisfaction levels of 78%, 68% and 77%. The tolerance ranks of these members are 5, 8 and 6, indicating that although DC 4 exhibits relatively lower satisfaction, the overall allocation remains within the acceptable tolerance threshold. For coalition {1, 9, 10}, the satisfaction levels of three DCs are 74%, 87% and 66%. The tolerance ranks of 3, 3 and 8 show that the allocation is still feasible. Coalition {5, 8} presents a feasible profit allocation outcome, with satisfaction levels of 100 percent and 50 percent for its two members. Although the satisfaction levels differ considerably, both data centers remain within their acceptable tolerance ranks of 1 and 5, indicating that the allocation is still consistent with the tolerance threshold.

Overall, although a few members show relatively low satisfaction values, all coalitions satisfy their tolerance constraints. The obtained matching results represent an optimal trade-off among coalition structure, profit allocation fairness, and total coalition profits.

To validate the effectiveness of the proposed rolling horizon control strategy, a case is conducted at $t = 30$, where operational strategies for the first 29 periods have already been implemented. At $t = 30$, only the strategies for periods 30 to 96 are re-optimized in response to uncertainty scenarios. Three optimal coalitions are simulated under three uncertainty scenarios, i.e., mild, moderate, and severe fluctuations, and the corresponding rolling horizon control results are presented in Table VI.

It can be seen from Table VI that under the mild fluctuation scenario, the coalition structure remains stable and the DCs continue to operate as the coalition {5, 8}. Only slight adjustments are required to compensate for small deviations in renewable generation from -2.0% to2.3%, and computational load from -2.4% to 2.6%. As a result, the net profit decreases slightly from 2903 CNY to 2902 CNY compared with the initial scenario, indicating that mild fluctuations bring only a marginal economic impact. Under the moderate fluctuation scenario, the coalition {2, 4, 6} also maintains its original structure. However, the forecast deviations become more significant, with renewable generation fluctuating from −6.0% to 5.7% and computational load fluctuating from −7.9% to 7.1%. This necessitates re-solving the collaborative operation model in Stage 2. As a result, the net profit decreases from 5114 CNY to 5030 CNY, suggesting that although the coalition configuration remains unchanged, operational recalibration is required to counteract the prediction errors. Under the severe fluctuation scenario, the coalition identified in the initial case cannot remain stable due to substantial deviations in renewable generation, ranging from −11.0% to 14.2%, and computational load, ranging from −14.5% to 14.6%. The coalition structure is re-optimized across Stages 1 to 3, resulting in two new coalitions, namely {1, 9} and {3, 7, 10}. Their total net profit reaches 9953 CNY, which is only 1.5% lower than the initial scenario value. This demonstrates that the proposed rolling horizon control strategy effectively mitigates the adverse effects of severe uncertainty fluctuations.

TABLE VI
ROLLING HORIZON CONTROL RESULTS UNDER THREE UNCERTAINTY SCENARIOS (t = 30-96)

Indicators	Initial scenario	Mild fluctuation	Moderate fluctuation	Severe fluctuation
Renewable generation fluctuation	/	-2.0%~2.3%	-6.0%~5.7%	-11.0%~14.2%
Computational load fluctuation	/	-2.4%~2.6%	-7.9%~7.1%	-14.5%~14.6%
Coalition structure	{3,7}, {2,4,6}, {1,9,10}, {5,8}	{5,8}	{2,4,6}	{1,9}, {3,7,10}
Net profit/CNY	3717, 5114, 6392, 2903	2902	5030	3887, 6066

TABLE VII
SOLUTION RESULTS COMPARISON BETWEEN M-WDA AND THE PROPOSED MODEL

Models	Coalition structure	Total profits/CNY	Net profit of each coalition/CNY	Net profit of each DC/CNY
M-WDA [35]	{3,7}, {2,4,6}, {1,9,10}, {5,8}	17701	3562, 5030, 6207, 2902	{1206,2356}, {1143,2093,1794}, {2016,1750,2441}, {1191,1711}
Proposed model	{1,9}, {2,4,6}, {3,7,10}, {5,8}	17885	3887, 5030, 6066, 2902	{2076,1811}, {1143,2093,1794}, {1207,2357,2503}, {1191,1711}

TABLE VIII
COMPUTATIONAL PERFORMANCE UNDER DIFFERENT DC SCALES AND MATCHING DEGREE THRESHOLDS

Indices	10	50			100		
		θ =0.9	θ =0.85	θ =0.8	θ =0.9	θ =0.85	θ =0.8
Potential coalition number	15	125	148	179	165	268	310
Serial solving time/h	4.9	38.2	>50	>50	>50	>80	>100
Parallel solving time (8 cores)/h	0.6	4.8	6.3	7.2	6.9	11.0	>12.5

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TABLE IX
MAIN TECHNICAL PARAMETERS OF THIS CASE

Parameters	Value	Parameters	Value	Parameters	Value
N/T	10/96	$u_{\mathrm{CPU}}^{i,\max}$	0.9	$F_{k,q}$	0.03-0.048 m^3/s
c_{air}	1.005	$C_{\mathrm{CPU}}^{\mathrm{Per},i}$	6 cores/GHz	$T_{\mathrm{DC,C}}^{\max}$ / $T_{\mathrm{DC,C}}^{\min}$	27/18
ρ_{air}	1.2 kg/m^3	$f_{\mathrm{CPU}}^{i,k,s}$	1.2-3.0 Hz	$V_{\mathrm{DC,C}}$ / $V_{\mathrm{DC,H}}$	120/100 m^3
M_k^{fan}	12	$C_{\mathrm{CPU}}^{\mathrm{Per},i}$	0.006 kW/GHz^2	$T_{\mathrm{DC,H}}^{\max}$ / $T_{\mathrm{DC,H}}^{\min}$	45/32
M_{ser}^{k}	72	$T_{\mathrm{b}}^{\max}$	30	ε_1 / ε_2	5% / 15%

To evaluate the advantages of the proposed optimal dynamic coalition matching mechanism based on rolling horizon control in Stage 3, it is compared with the optimal coalition matching mechanism without dynamic adjustments (M-WDA) [35]. The comparison results are presented in Table VII. It can be seen from Table VII that under severe source–load fluctuations, the M-WDA model keeps the coalition structures {3, 7} and {1, 9, 10} unchanged. As a result, these two coalitions obtain total profits of 3562 CNY and 6207 CNY, respectively, yielding a combined profit of 9769 CNY. In contrast, the proposed model is able to re-execute the optimization procedure of Stages 1 to 3 when severe deviations occur, enabling the original coalitions to be reorganized into the more compatible structures {1, 9} and {3, 7, 10}. The profits of these newly formed coalitions increase to 3887 CNY and 6066 CNY, resulting in a combined total of 9953 CNY, which is 1.9% higher than that of M-WDA. As a result, the total profit of the proposed method is 1.1% higher than that of the M-WDA. This improvement illustrates that the proposed model can adaptively reconstruct coalition structures in response to large disturbances, thereby enhancing both economic efficiency and coalition stability under uncertainty.

D. Scalability Analysis of the Proposed Method

To further evaluate the scalability and practical applicability of the proposed triple-stage optimization method, Table VIII presents the computational performance under different DC scales and various matching degree thresholds. It can be seen from Table VIII that the solving time increases with the number of DCs, primarily because Stage 2 requires solving a collaborative optimization model and performing Shapley-value-based profit allocation for each potential coalition. For the case of $N = 10$, the serial solving time is approximately 4.9 h, whereas 8-core parallel computing reduces it to only 0.6 h. As the DC scale increases, the number of potential coalitions grows substantially (e.g., 125-179 for $N = 50$, and 268-310 for $N = 100$), resulting in longer serial solving times that may exceed 100 h. However, because Stage 2 optimization problems for different potential coalitions are mutually independent, parallel computation can significantly reduce the overall runtime. With 8-core parallelization, the solving time decreases to 4.8-7.2 h for $N = 50$ and 6.9-12.5 h for $N = 100$, representing more than an 80% reduction compared with serial execution.

These results demonstrate that the proposed method maintains strong scalability for large-scale cross-regional DC systems. The matching degree threshold can also be appropriately adjusted to reduce the number and size of potential coalitions, enabling further computational savings with negligible impact on solution quality. Moreover, since Stage 2 optimization is inherently parallelizable, modern multi-core computational resources allow the framework to efficiently support larger DC networks while maintaining acceptable solving times. Overall, the proposed triple-stage optimization method is computationally tractable and well-suited for practical large-scale deployment.

VII. CONCLUSION

A triple-stage collaborative optimization method for cross-regional DCs considering optimal coalition matching is proposed in this paper. Case studies on ten cross-regional DCs with heterogeneous resource characteristics were conducted to evaluate the performance of the proposed method, and the following conclusions can be drawn:

1) The proposed potential coalition search method in Stage 1 efficiently identifies high-quality and structurally stable coalitions. Compared with the conventional Bron–Kerbosch algorithm, the proposed method produces identical maximal clique results while reducing recursion count and computation time by up to 22.98% and 18.74% in large-scale cases, demonstrating superior scalability and computational efficiency.

2) The proposed optimal collaborative operation model in Stage 2 features fine-grained cooling system modeling that accurately captures multiple physical characteristics, while jointly optimizing computational load scheduling and energy management. Compared with other baseline counterparts, it reduces total energy cost by at least 3.4%, lowers cooling energy consumption by at least 2.1%, and improves PUE to 1.59, enhancing both energy efficiency and economic performance.

3) The proposed optimal dynamic coalition matching mechanism in Stage 3 balances coalition-wide profit and member satisfaction by integrating satisfaction and tolerance thresholds. Under severe fluctuation scenarios, the rolling-horizon control further enhances adaptability, raising total profit by 1.1% compared with M-WDA.

In future work, privacy-preserving distributed optimization methods will be explored to further reduce data-sharing requirements within coalitions and enhance the practicality of the proposed framework in large-scale DC deployments.

APPENDIX

See Fig. 12 and Table IX.

REFERENCES

[1] T.Han, Y. Wang, Z. Mi, K. Han, J. Tian, and Y.-M. Wei, "Designing and regulating clean energy data centres," *Nature Rev. Clean Technol.*, vol. 1, no. 6, pp. 373–374, May 2025.

[2] E. Masanet, A. Shehabi, N. Lei, S. Smith, and J. Koomey, "Recalibrating global data center energy-use estimates," *Science*, vol. 367, no. 6481, pp. 984–986, Feb. 2020.

[3] C. Lin, Y. Xu, Z. Yi, and Z. Tu, "Energy management for rural microgrid with inaccurate equipment parameters: A KAN-based deep reinforcement learning method," *IEEE Trans. Smart Grid*, vol. 16, no. 6, pp. 4808–4821, Nov. 2025.

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[4] Y. Liu, X. Wei, J. Xiao, Z. Liu, Y. Xu, and Y. Tian, "Energy consumption and emission mitigation prediction based on data center traffic and PUE for global data centers," *Global Energy Interconnection*, vol. 3, no. 3, pp. 272–282, Jun. 2020.
[5] C. Jin, X. Bai, C. Yang, W. Mao, and X. Xu, "A review of power consumption models of servers in data centers," *Appl. Energy*, vol. 265, May 2020, Art. no. 114806.
[6] S. M. Nabavinejad, S. Reda, and M. Ebrahimi, "Coordinated batching and DVFS for DNN inference on GPU accelerators," *IEEE Trans. Parallel Distrib. Syst.*, vol. 33, no. 10, pp. 2496–2508, Oct. 2022.
[7] Y. Zhou et al., "Energy savings in direct air-side free cooling data centers: A cross-system modeling and optimization framework," *Energy Buildings*, vol. 308, Apr. 2024, Art. no. 114003.
[8] Y. Zhou, F. Wei, S. Li, Z. Wang, J. Liu, and D. Yu, "Data center load modeling through optimal energy consumption characteristics: A path to simultaneously enhance energy efficiency and demand response quality," *Appl. Energy*, vol. 393, Sep. 2025, Art. no. 126095.
[9] J. Wang, H. Deng, Y. Liu, Z. Guo, and Y. Wang, "Coordinated optimal scheduling of integrated energy system for data center based on computing load shifting," *Energy*, vol. 267, Mar. 2023, Art. no. 126585.
[10] J. Han, K. Han, T. Han, Y. Wang, Y. Han, and J. Lin, "Data-driven distributionally robust optimization of low-carbon data center energy systems considering multi-task response and renewable energy uncertainty," *J. Building Eng.*, vol. 102, May 2025, Art. no. 111937.
[11] H. Guo, H. Yu, M. Wang, C. Liu, and C. Li, "Integrated management of workloads and energy system for data centers," *Energy*, vol. 327, Jul. 2025, Art. no. 136400.
[12] Z. Ding et al., "Data center job scheduling and energy management under uncertain environments," *IEEE Trans. Ind. Appl.*, vol. 61, no. 4, pp. 5489–5500, Jul. 2025.
[13] B. Fatemipour, Z. Zhang, and M. St-Hilaire, "A survey on replica transfer optimization schemes in geographically distributed data centers," *IEEE Trans. Netw. Service Manage.*, vol. 21, no. 6, pp. 6301–6317, Dec. 2024.
[14] J.-W. Xiao, Y.-B. Yang, S. Cui, and Y.-W. Wang, "Cooperative online schedule of interconnected data center microgrids with shared energy storage," *Energy*, vol. 285, Dec. 2023, Art. no. 129522.
[15] L. Yang, X. Chen, X. Fang, and Q. Yang, "Optimal coordinated management of integrated electricity-heat-computation systems in geographically distributed data centers," *Energy*, vol. 332, Sep. 2025, Art. no. 137232.
[16] S. Zhou, M. Zhou, Z. Wu, Y. Wang, and G. Li, "Energy-aware coordinated operation strategy of geographically distributed data centers," *Int. J. Electr. Power Energy Syst.*, vol. 159, Aug. 2024, Art. no. 110032.
[17] Y. Sun, Z. Ding, Y. Yan, Z. Wang, P. Dehghanian, and W.-J. Lee, "Privacy-preserving energy sharing among cloud service providers via collaborative job scheduling," *IEEE Trans. Smart Grid*, vol. 16, no. 2, pp. 1168–1180, Mar. 2025.
[18] J. Sun, Y. Yan, Z. Wang, J. Ma, and Y. Wang, "Privacy-preserving coordinated operation of cross-enterprise data centers," *Appl. Energy*, vol. 383, Apr. 2025, Art. no. 125343.
[19] C. Chen et al., "Cooperative-game-based day-ahead scheduling of local integrated energy systems with shared energy storage," *IEEE Trans. Sustain. Energy*, vol. 13, no. 4, pp. 1994–2011, Oct. 2022.
[20] R. Massin, C. J. Le Martret, and P. Ciblat, "A coalition formation game for distributed node clustering in mobile ad hoc networks," *IEEE Trans. Wireless Commun.*, vol. 16, no. 6, pp. 3940–3952, Jun. 2017.
[21] J. Han et al., "Evaluating the dispatchable capacity of cross-regional data center clusters toward power system operation," *IEEE Trans. Smart Grid*, early access, Oct. 2025, doi: 10.1109/TSG.2025.3623107.
[22] T. Qin, M. Liu, S. Ji, and D. Cai, "Parameter weight analysis of synchronous induction electromagnetic coil launch system based on the entropy weight method," *IEEE Trans. Plasma Sci.*, vol. 52, no. 5, pp. 1865–1873, May 2024.
[23] K. Petroutsatou, I. Ladopoulos, and D. Nalmpantis, "Hierarchizing the criteria of construction equipment procurement decision using the AHP method," *IEEE Trans. Eng. Manag.*, vol. 70, no. 9, pp. 3271–3282, Sep. 2023.
[24] S. Zhang, J. Lyu, W. Jin, H. Cheng, C. Li, and X. Wang, "Coordinated planning of multiple energy hubs considering the spatiotemporal load regulation of data centers," *IEEE Trans. Power Syst.*, vol. 39, no. 2, pp. 4193–4207, Mar. 2024.
[25] C. Guo, F. Luo, Z. Cai, and Z. Y. Dong, "Energy management of internet data centers in multiple local energy markets," *Electr. Power Syst. Res.*, vol. 205, Apr. 2022, Art. no. 107760.
[26] L. Yu, T. Jiang, Y. Zou, and Z. Sun, "Joint energy management strategy for geo-distributed data centers and electric vehicles in smart grid environment," *IEEE Trans. Smart Grid*, vol. 7, no. 5, pp. 2378–2392, Sep. 2016.
[27] Y. Qu and J. Lee, "On disjunction convex hulls by big-M lifting," *Discrete Appl. Math.*, vol. 371, pp. 31–45, Aug. 2025.
[28] C. Chen et al., "Triple-level full-stage adaptive restoration optimization for coupled transmission and distribution systems considering restoration security risks," *IEEE Trans. Power Syst.*, vol. 39, no. 6, pp. 6833–6848, Nov. 2024.
[29] Alibaba Group E-Commer. Co. (Nov. 25, 2018). *Alibaba Cluster Trace Program[EB/OL]*. Accessed: Sep. 8, 2025. [Online]. Available: https://github.com/alibaba/clusterdata/tree/v2018/cluster-trace-v2018
[30] Z. Wang et al., "Distributionally robust planning for data center park considering operational economy and reliability," *Energy*, vol. 290, Mar. 2024, Art. no. 130185.
[31] W. Chen, P. Hao, Z. Wei, and L. Chen, "Study on the optimization allocation method of distributed energy storage in an active distribution network taking into account transmission betweenness and source-network-load synergy," *Electr. Power Syst. Res.*, vol. 247, Oct. 2025, Art. no. 111787.
[32] Y.-W. Wei, W.-M. Chen, and H.-H. Tsai, "Accelerating the Bron–Kerbosch algorithm for maximal clique enumeration using GPUs," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 9, pp. 2352–2366, Sep. 2021.
[33] R. Tao, D. Zhao, C. Xu, H. Wang, and X. Xia, "Resilience enhancement of integrated electricity-gas-heat urban energy system with data centres considering waste heat reuse," *IEEE Trans. Smart Grid*, vol. 14, no. 1, pp. 183–198, Jan. 2023.
[34] J. Wang, W. Deng, C. Yue, W. Su, and X. Bai, "Joint optimization of cooling parameters and workload distributions based on model predictive control for rack-based data centers," *J. Building Eng.*, vol. 100, Apr. 2025, Art. no. 111801.
[35] Q. Huang, Y. Liu, L. Wang, P. Sun, J. Li, and J. Xu, "A blockchain-enabled coalitional game framework for carbon emission trading," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 6, pp. 5503–5514, Nov. 2024.



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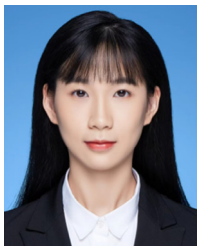
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